PERFORMANCE ANALYSIS OF RIS AIDED WIRELESS SYSTEMS AND JOINT TRANSMISSION IN IEEE 802.11be

A THESIS

submitted by

Ashutosh Soni

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in

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DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

INDIAN INSTITUTE OF TECHNOLOGY BHILAI

MAY 2022

Dedicated to my Mom, Dad, Sister and friends.

DECLARATION BY THE STUDENT

This is to certify that the thesis titled **PERFORMANCE ANALYSIS OF RIS AIDED WIRELESS SYSTEMS AND JOINT TRANSMISSION IN IEEE 802.11be** has been authored by me. It presents the research conducted by me under the supervision of **Dr. Sreejith T. V.** and **Dr. Arzad Alam Kherani**. To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in full or parts, for the award of a degree or diploma. Further, due credit has been attributed to the relevant state-of-the-art and collaborations (if any) with appropriate citations and acknowledgements, in line with established norms and practices.



Place: Raipur
Date: 9th May 2022

Ashutosh Soni
Bachelor of Technology (Honours)
Electrical Engineering
Indian Institute of Technology Bhilai

CERTIFICATE AGAINST PLAGIARISM

This is to certify that the thesis entitled **PERFORMANCE ANALYSIS OF RIS AIDED WIRELESS SYSTEMS AND JOINT TRANSMISSION IN IEEE 802.11be**, submitted by **Ashutosh Soni**, a bonafide record of the research work, done under my supervision is submitted to Indian Institute of Technology Bhilai on **April 22, 2022** to meet the requirements for the award of **Bachelor of Technology (Honours)** degree.

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Place: Raipur
Date: 9th May 2022

Dr. Sreejith T. V.
Assistant Professor
Department of Electrical Engineering
and Computer Science
Indian Institute of Technology Bhilai



Dr. Arzad Alam Kherani
Associate Professor
Department of Electrical Engineering
and Computer Science
Indian Institute of Technology Bhilai

I have personally verified the report and find no significant similarity in the thesis.



Place: Raipur
Date: 9th May 2022

Dr. Prashant Agnihotri
Convener, DUGC
Department of Electrical Engineering
and Computer Science
Indian Institute of Technology Bhilai

THESIS CERTIFICATE

This is to certify that the thesis entitled **PERFORMANCE ANALYSIS OF RIS AIDED WIRELESS SYSTEMS AND JOINT TRANSMISSION IN IEEE 802.11be**, submitted by **Ashutosh Soni** to Indian Institute of Technology Bhilai, for the award of the degree of **Bachelor of Technology (Honours)** is a bonafide record of research work carried out by the student under my supervision. The contents of the thesis, in full or in parts, have not been separately submitted to any Institute or University for the award of a degree or diploma.



Place: Raipur
Date: 9th May 2022

Dr. Sreejith T. V.
Assistant Professor
Department of Electrical Engineering
and Computer Science
Indian Institute of Technology Bhilai



Dr. Arzad Alam Kherani
Associate Professor
Department of Electrical Engineering
and Computer Science
Indian Institute of Technology Bhilai

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ABSTRACT

Keywords : *Re-configurable Intelligent Surfaces*, *SimRIS*, *Wi-Fi 7*, *Multi-AP Coordination*, *Joint Transmission*, *Queue Analysis*

With the increase in demand for higher datarates at a very low latency increases to support numerous different use cases as we move towards data-driven and data-intensive world, the need to investigate ways to improve the existing communication infrastructure is cruicial. In this work, we have tried to analyse two different domains of active research as of today namely - Re-configurable Intelligent Surfaces and Multi-AP Coordination in WiFi. With the emergence of Smart Radio Environments (SREs), the ability to control the signal propagation is very helpful to utilise and virtually mould the environment to our use. Reconfigurable Intelligent Surfaces can be used to assist exsiting communication systems to improve the SNR for cell edge users. Our work provides a new way to incorporate RIS panels in traditional communication system and analyse the behavior of received SNR versus system parameters. We also present insights that there will exist regions where the relative benefit per RIS deployed is better than other regions, which could be exploited in RIS allocation. The second problem this work tries to analyse is to serve users using coordinated transmission such that the user gets the same signal in a constructive manner from multiple transmitters. This is an active research in the Wi-Fi 7 standardisation process. We present results that suggests the concept is beneficial in Wi-Fi networks. Further we present a scheduling scheme named fluid model for different modes of service, explaining the queue evolution, stability and conditions for stationary queue.

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ABBREVIATIONS

3GPP	3rd Generation Partnership Project
BS	Base Station
UE	User Equipment
UAV	Unmanned Aerial Vehicle
RIS	Re-configurable Intelligent Surfaces
SRE	Smart Radio Environment
SNR	Signal to Noise Ratio
SINR	Signal to Interference plus Noise Ratio
AP	Access Point
STA	Station
SAPT	Single Access Point Transmission
SAPI	Single Access Point transmission with Interference
JT	Joint Transmission

CHAPTER 1

Introduction - Re-configurable Intelligent Surfaces Assisted Networks

According to Cisco's February 2019 research the number of networked devices and connections will reach 28.5 billions by 2022, with 12.3 billions of those being mobile-ready devices and connections. In addition, by 2022, total mobile data traffic is estimated to reach 77 exabytes per month, a seven-fold increase over 2017. Meanwhile, in June 2018, the first commercial 5th generation (5G) mobile communication standard (3GPP Release 15) was finalised after years of study and development. 5G wireless networks have already been installed in several countries as of mid-2019, and the first 5G compatible mobile handsets are on the market. The introduction of 5G has ushered in a new era of mobile communications, one that includes three distinct use cases with distinct requirements: increased mobile broadband, ultra-reliable and low-latency communications, and huge machine-to-machine interactions. During the standardisation process for 5G wireless networks, one thing has become clear: no one enabling technology can fulfil all 5G application needs. In this light, researchers have already begun work on technologies beyond 5G, or even 6th generation (6G), by breaking out from the confines of 5G-centric solutions.

It is very obvious that, in order to meet the ever-increasing need for data, any advancement in existing technology is beneficial. Better optimization may lead to improved spectral efficiency, with the radio spectrum being one of the most essential resources. Nonetheless, we must consider new paradigms in communication systems that have the potential to alter our understanding of communication. One of many such study projects is the use of Re-configurable Intelligent Surfaces (RISs) to help current wireless networks by boosting received SNR and hence user capacity.

1.1 Background

1.1.1 Re-configurable Intelligent Surfaces

Re-configurable Intelligent Surfaces is a new revolutionary technology proposed in [13] that aims to improve the spectrum and energy efficiency of wireless communications, proposing a low hardware cost solution. The RISs are man-made planar surfaces of electromagnetic (EM) material consisting of large number of passive elements that can be controlled electronically with electronic circuitry such as micro-controllers. These surfaces have unique wireless communication capabilities in terms that each element in the surface is able to introduce a phase difference in the received signal in an independent manner from the other such elements on the surface. This behaviour of introducing phase shifts helps to redirect the incident signal to a desired direction as the phase differences from several elements will add up constructively in certain directions and will cancel out in other. These surfaces can therefore achieve three-dimensional (3D) passive beamforming without requiring radio-frequency (RF) chains. An example of RIS assiting multi-user communication system is shown in figure 1.1. We can clearly see the different User Equipments (UEs) receive signals from both the LoS and the through the reflected path through the RIS.

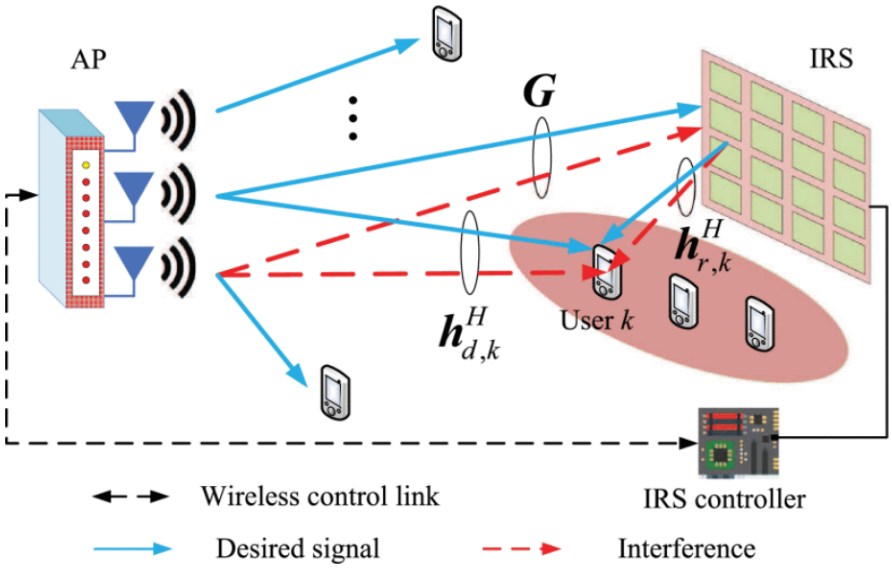


Figure 1.1: RIS aided multiuser communication illustration [13]

There are various other existing technologies in the literature that works to provide similar support to improve communication quality. Some of these are amplify-and-forward (AF) relay, back-scatter communication and active intelligent surface based massive multiple-input-

multiple-output (MIMO). A comparison between RIS with the aforementioned technologies is shown in figure 1.2. It is noteworthy that the RIS does not require a transmitter module as it just reflects the incident signals. Due to its passive nature, the reflection part does not consume any power, though power is required to drive the micro-controller controlling the phase shifts for each element. It can work in full-duplex mode i.e. it can support communication in both the directions at the same time. The hardware being in the form of a panel provides advantage in terms of flexibility to place it on the ceiling/walls as per the requirement.

Technology	Operating mechanism	Duplex	No. of transmit RF chains needed	Hardware cost	Energy consumption	Role
IRS	Passive, reflect	Full duplex	0	Low	Low	Helper
Backscatter	Passive, reflect	Full duplex	0	Very low	Very low	Source
MIMO relay	Active, receive and transmit	Half/full duplex	N	High	High	Helper
Massive MIMO	Active, transmit/receive	Half/full duplex	N	Very high	Very high	Source/ Destination

Figure 1.2: Comparing RIS with other similar technologies [13]

Current implementations include conventional reflect-arrays, liquid crystal surfaces, and software-defined meta-surfaces[10, 8, 11].

1.1.2 Related Work

The analysis on RIS-aided single-cell wireless system where one RIS is deployed to assist in the communications between a multi-antenna access point (AP) and multiple single-antenna users is presented in [13]. The authors have tried to formulate and present a solution for power minimisation at the transmitter by jointly optimising the active beamforming and passive beamforming at the transmitter and the RIS respectively. The paper claims that they could achieve the same rate performance as massive MIMO that does not use RIS implying the usefulness of the RIS assisted networks. The research in [14] is widely popular as it discusses the key challenges in designing a RIS assisted network setup. It presents results which helps them conclude that the use of RIS in traditional wireless networks provides significant improvements and they foresee that such an integration of active and passive components in communication systems will fundamentally change the architecture.

Authors in [3] analysed the improvements of using software-controlled RIS over the classic decode-and-forward (DF) relaying technique. The concept of relaying is often confused with the idea of RIS, but the way the two work is a lot different. The paper concludes that in order to outperform DF relaying on the parameters of minimising the total transmit power and energy

efficiency, very high rates or RIS panel of very large sizes are required. On similar lines, the paper [4] investigates the efficiency of RIS-assisted wireless system in terms of average SNR, outage probability (OP), symbol error rate (SER) etc. and compared it with a standard amplify-and-forward (AF) relay. The results indicate that the use of RIS clearly outperforms the corresponding AF-relayed systems in terms of the parameters mentioned before.

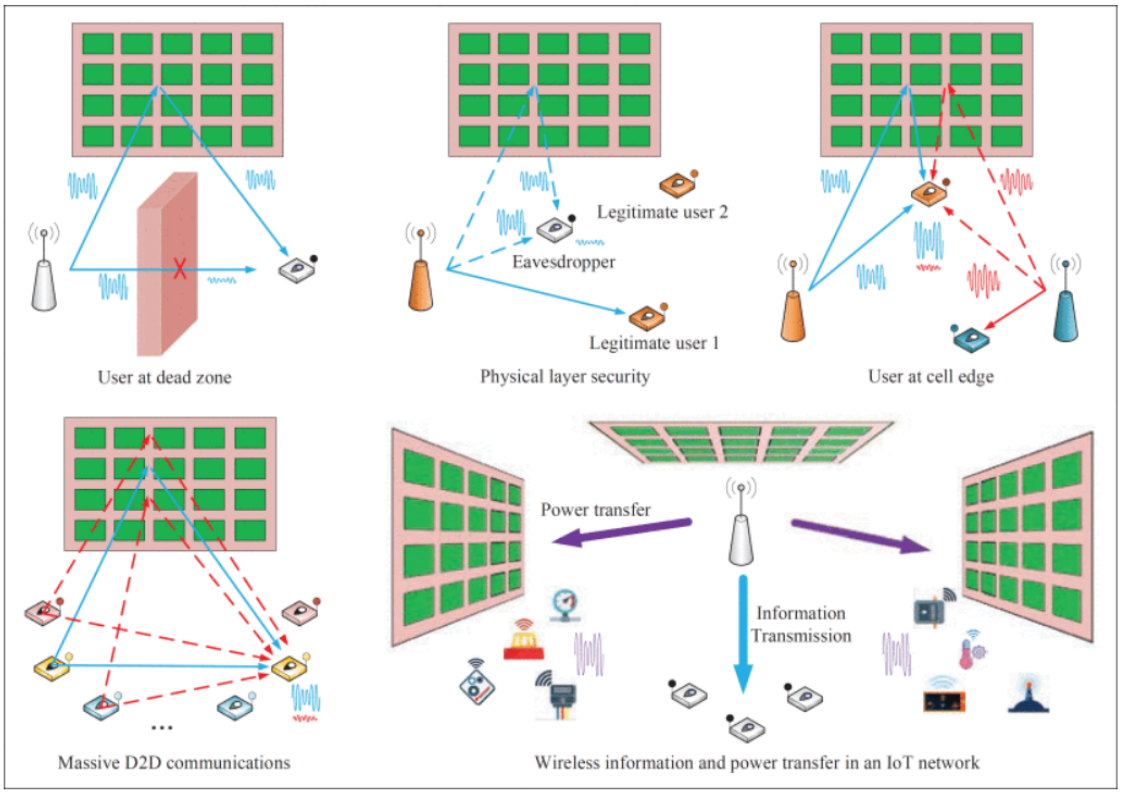


Figure 1.3: Use cases of RIS [13]

1.1.3 Use Cases

The concept of being able to influence the radio environment in a controlled manner to achieve better communication is helpful in a lot of different use cases like:

1. In case where the LoS path is blocked and we need a controlled indirect link between the Tx and Rx.

2. If the LoS link is not strong enough to support the Quality of Service (QoS) requirements of the UE, the power received through the RIS path can help improve the SNR helping meet the QoS requirement.

3. In an IoT network which consist of a large number of low-power devices, it is likely that the devices cannot transmit to large distances. In such a case, RIS panels can act like a signal reflector hub assisting Device-to-Device (D2D) communication, utilising the reflected signals through walls and ceilings that in general are not put to use.

The analysis in the thesis primarily deals with the second use case, analysing the received SNR for stationary and mobility scenarios.

1.2 Organisation of Thesis for Re-configurable Intelligent Surfaces Assisted Networks

The part of the thesis explaining the research on RIS-assisted networks in noise limited scenario is divided in 2 chapters namely - Introduction and Uplink SNR Analysis in Noise Limited Scenario. The chapter on introduction focuses on motivation for the topic, background information on RIS, related work and use cases. The chapter on uplink SNR analysis focuses on understanding SNR variations with system parameters in stationary and mobile scenarios. Some insights on using the plots for RIS allocation are also mentioned.

CHAPTER 2

Uplink SNR Analysis in Noise Limited Scenario

The analysis for uplink data transmission that is presented and explained in this thesis has been published in the paper [6].

2.1 System Model

The system model consists of base stations (BSs) and reconfigurable intelligent surfaces (RISs) that are spatially distributed on an infinite plane according to independent 2-dimensional Poisson processes of intensity β and ρ, respectively. There is an unmanned aerial vehicle (UAV), that acts like a user equipment (UE) in this scenario, which connects and communicates with the nearest BS. The signals for uplink data transfer from the UAV to the BS can go through the Line of Sight (LoS) path and the Non-Line of Sight (NLos) path. We assume simple path loss model for signal propagation and power calculations in the system with path loss exponent α.

The communication through NLoS will be assisted by the RIS elements as they will redirect the signals in the direction of the correct BS the UAV is currently connected to. We assume that this reflection towards a particular location is possible as per the existing research in several papers and do not go in the details of the inner functioning of the RIS itself and the way it achieves directional redirection. The different signals coming after reflection from different RIS panels will have phase differences due to which they might not add constructively at the BS. Since the RIS can control the phase of the signal during reflection, we assume that all the signals through the RIS panels will add constructively.

For a UAV-BS pair which are communicating at a particular time, we assume a sector around the BS characterised by sector angle $2\bar{\theta}$ and sector radius r. Since there will be infinite number of RIS elements on an infinite plane, the sector we defined helps us focus on a finite amount of RIS for analysis. Only the effect of RIS that fall within this sector will be considered for SNR analysis and the effect of RIS outside the sector will be neglected. In a sense, this is a way to say that the RIS that fall in the sector are allocated to this UAV-BS pair. Since only the effect of RIS in the sector is captured, the sector angle and sector radius will be termed as *capture angle* and *capture radius*. Let us denote the distance between the UAV and BS as r_{LoS}. The system

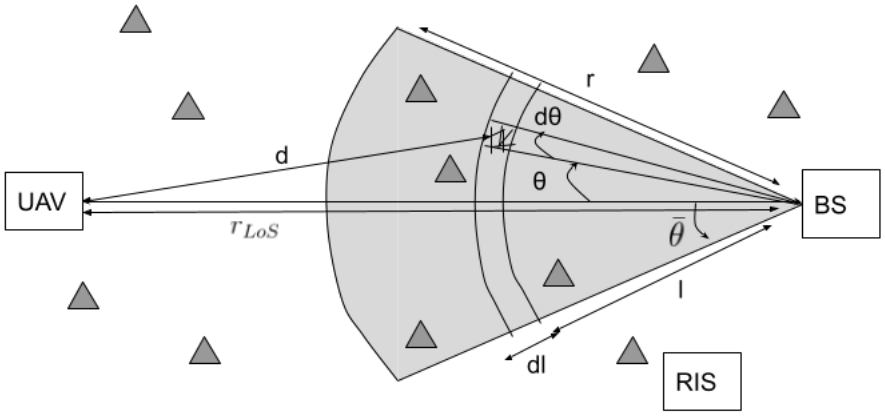


Figure 2.1: Illustration of system model for a use case of UAV systems. The capture area is shaded.

model is shown in Fig 2.1.

Remark 2.1.1 *We assume that the changes in the altitude of the UAV is small enough to consider that it remains in the plane of BS and RIS throughout the communications.*

2.2 Power calculations

For the system model explained in the previous section, we are interested to perform Signal-to-Noise Ratio (SNR) analysis and understand the effect on the SNR values as the system parameters changes. To obtain the SNR values, we first need to characterise the received power for the signal transmitted by the UAV to the BS.

Definition 2.2.1 ***Path loss model****: this is a model which assumes that theere is a line of sight (LoS) path and a non-lline of sight (NLoS) path (due to presence of scatterers) for the signal to reach the receiver. If the transmitted and received signal power is represented by P_t and P_r respectively, then the relation between the two is given by:*

$$P_r = D_t D_r \left(\frac{\lambda}{4\pi}\right)^2 \frac{P_t}{d^\alpha} \tag{2.1}$$

where α represents the path loss exponenet. $\alpha = 2$ when there are no scatterers i.e. free space path loss model.

Remark 2.2.1 *We will assume $D_t = 1$ and $D_r = 1$ for the calculations and analysis throughout.*

The received signal power is a combination of the power of the conventional signal (combination of a direct and indirect path), denoted by P_D, and the power through various RIS denoted as P_R. We assume that the there are no other reflecting surfaces in the system due to which we can consider the only conventional signal is the signal from the LoS path. The power of the conventional signal, P_D, using equation 2.1 is given by,

$$P_D = \left(\frac{\lambda}{4\pi}\right)^2 \frac{P_t}{r_{LoS}^{\alpha}} \tag{2.2}$$

where λ is the wavelength used.

We assume a free path loss model for the signal received through the RIS path i.e. $\alpha = 2$. Lets assume there are M RIS elements in the $(r, 2\bar{\theta})$-sector for a UAV-BS pair, then using the figure 2.1 the aggregate average received power from all the paths through RIS can be written as as product of the attenuation in the UAV-RIS path and the RIS-BS paths denoted by $A_{UAV-RIS}$ and A_{RIS-BS}:

$$P_R = \left[\sum_{i=1}^{M} P_t * A_{UAV-RIS} * A_{RIS-BS}\right] \tag{2.3}$$

$$= \left[\sum_{i=1}^{M} \left(\left(\frac{\lambda}{4\pi}\right)^2 \frac{P_t}{{d_i}^2}\right) * \left(\left(\frac{\lambda}{4\pi}\right)^2 \frac{1}{{l_i}^2}\right)\right] \tag{2.4}$$

$$= \left[\sum_{i=1}^{M} \frac{\lambda^4 P_t}{(4\pi)^4 d_i^2 l_i^2}\right] \tag{2.5}$$

here, the expectation is with respect to M, where M is Poisson distributed with mean $2\rho r^2\bar{\theta}$. d_i and l_i represent the distances from UAV to the RIS and RIS to the BS respectively, P_t is the transmitted signal power.

2.3 Noise Limited Scenario

Let us consider the uplink transmission in a noise-limited scenario i.e. the interference in the system is taken to be zero. In such a system, the received SNR will be a function of received power through direct LoS path (P_D), received power through RIS panels (P_R) and the noise denoted by P_N. Equation 2.6 represents the received SNR, denoted by S_r, in mathematical form.

$$S_r = \frac{(P_D + P_R)}{P_N} \tag{2.6}$$

In equation 2.5 we can observe that M depends on $\bar{\theta}, r$ and ρ. Since the RIS are spread using a Poisson point process which is a random process, the expectation in the equation can be found out using integration.

Using the polar coordinate system centered at the connected base station to represent the possible RIS locations, an expression for P_R is obtained by identifying that the probability of presence of RIS in a small area $l(dl)(d\theta)$ around the point (l, θ) is $\rho l(dl)(d\theta)$ (see Figure 2.1). The expression for P_R is represented in form of a function dependent on the variables $r_{LoS}, r, \bar{\theta}$ and ρ as represented in equation 2.7.

$$P_R(r_{LoS}, r, \bar{\theta}, \rho) = P_t N \left(\frac{\lambda}{4\pi}\right)^4 * \int_{-\bar{\theta}}^{\bar{\theta}} \int_1^r (d(l, \theta, r_{LoS}) * l)^{-2} \rho l(dl)(d\theta) \tag{2.7}$$

where N denotes the number of individual RIS elements present in a RIS panel. $d(l, \theta, r_{LOS})$ denotes the variable path length from the UAV to the RIS element located at distance l from the BS making an angle θ from the straight line connecting the UAV and the BS. Using figure 2.2 This distance can be calculated as:

$$d(l, \theta, r_{LOS}) = \sqrt{(r_{LOS} - l\cos\theta)^2 + (l\sin\theta)^2}. \tag{2.8}$$

Remark 2.3.1 *We can see that the integration limit for r in equation 2.7 starts from 1 rather than 0. As the distance between the UAV-BS pair decreases to less than 1, the corresponding received power explodes due to nature of fraction numbers. The limit starts from 1 in order to avoid the near field issues with using Friss Equation.*

2.4 Analysis assuming $\alpha = 2$ for direct UAV-BS link

Let us assume free space path loss model for the direct link between the UAV and BS for signal propagation i.e. $\alpha = 2$. In a stationary scenario where the UAV is stationary i.e. does not move around, we can analyse the changes in received SNR, S_r, with changes in r and $\bar{\theta}$. The simulations for the analysis is carried out using SimRIS channel simulator [2] and the theoritical plots are obtained using Python programming. The values of P_R cannot be found

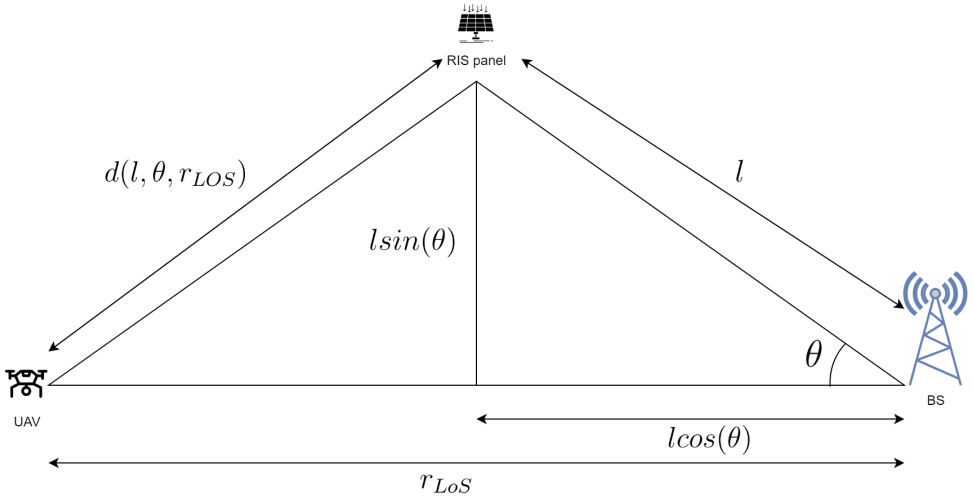


Figure 2.2: Calculation of the distance from the UAV to a RIS at location (l,θ)

in form of a closed loop formula from equation 2.7 and is therefor found out using numerical integration techniques using SciPy packages in Python.

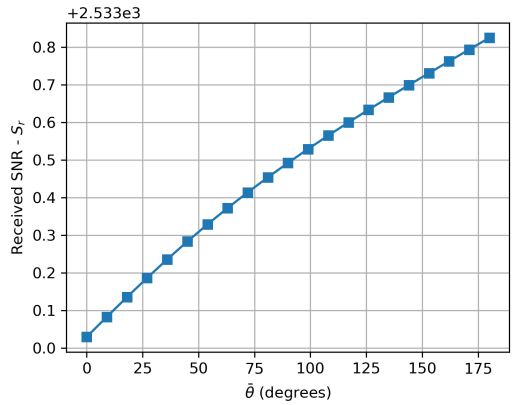
Variables used in theoretical results plotting are given in table 2.1. It is to be noted that these are the values used when these variables are treated as constants in a plot, in case of a plot vs one of the variables, the range of variation is also given.

Parameter	**Value**
P_t	30 dBm (1 W)
P_N	-100 dBm (10^{-10} mW)
r	50 (m)
r_{LoS}	100 (m)
$\bar{\theta}$	60
ρ	0.01 (per sq. m)
β	0.00015625 (per sq. m)
λ	1 cm (30 Ghz)

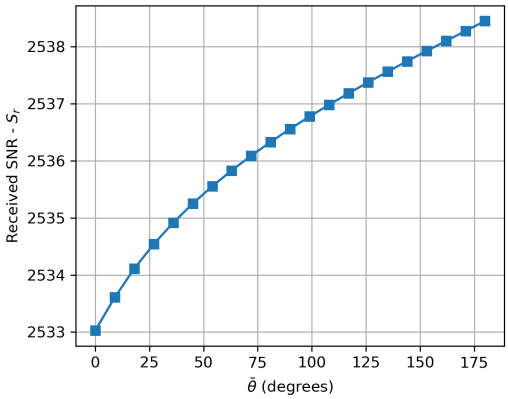
Table 2.1: Parameters used for theoretical plots

The theoretical plots for received SNR, S_r, versus the variation in capture angle $\bar{\theta}$ is shown in figure 2.3. We can see that as $\bar{\theta}$ varies from 0 degrees to 180 degrees, the variation in the value of S_r is in the first decimal while the values are around 2533 (the SNR is high due to lesser UAV-BS distance and lesser attenuation due to free space path loss). The variation, though small, can be seen to be linear indicating linear relationship between received SNR and capture angle.

Similarly, the theoretical plots for received SNR, S_r, versus the variation in capture radius r is shown in figure 2.4. From the plots we can observe that the variation in the received SNR as r goes from 1m to 150m in figure 2.4a and from 1m to 250m in figure 2.4b is around 0.042% and 0.118 % respectively. Though the variation is negligible, we can observe that the plot has



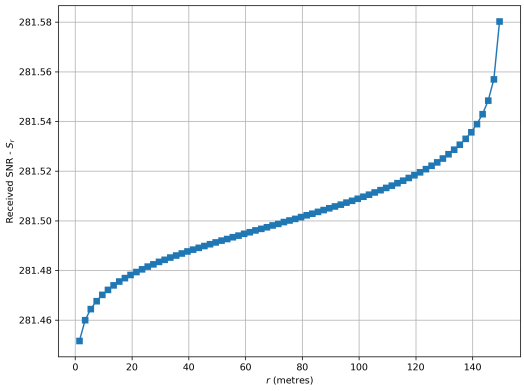
(a) S_r Vs $\bar{\theta}$ for $r = 20$m, $r_{LoS} = 50$m, $\rho = 0.1$ per sq. m



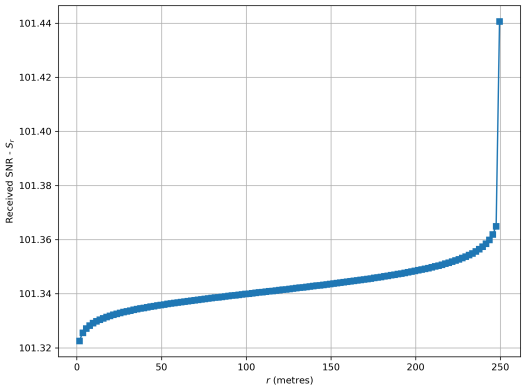
(b) S_r Vs $\bar{\theta}$ for $r = 40$m, $r_{LoS} = 50$m, $\rho = 0.5$ per sq. m

Figure 2.3: Received SNR S_r Vs capture angle $\bar{\theta}$ plot $\alpha = 2$

a characteristic shape where at the beginning and towards the end, we can observe the rate of change of S_r is higher than the middle region.



(a) S_r Vs r for $\bar{\theta} = 60°$, $r_{LoS} = 150$ m, $\rho = 0.1$ per sq. m



(b) S_r Vs r for $\bar{\theta} = 60°, r_{LoS} = 250$ m, $\rho = 0.1$ per sq. m

Figure 2.4: Received SNR S_r Vs capture radius r plot for $\alpha = 2$

2.4.1 Simulation Setup

The simulation is done using SimRIS simulator [2], which is an open-source and MATLAB-based simulator, which can be used in channel modeling of RIS-assisted MIMO systems with tunable operating frequency, terminal locations, number of RIS elements, and environments by performing Monte Carlo simulations at 28 and 73 GHz frequencies. We considered a square grid of size 250m*250m with each node in the grid representing a possible position of the RIS panel. The LoS channels with $\alpha = 2$ is stored for each such point using SimRIS simulator. Depending on the values of r and $\bar{\theta}$, we can add the contributions from the RIS panels falling in the sector around the BS. This value will be multiplied with the transmit power P_t and intensity

(or rate) of the poisson distribution of the RIS panels denoted by ρ. The value obtained after such a simulation will be close to the expected value of the received SNR if the RIS panels were randomly distributed during the simulation.

The simulation plots for variation in received SNR, S_r, as $\bar{\theta}$ and r increases in shown in figure 2.5 and 2.5 respectively. The plots show the contributions from the LoS and RIS in the total SNR.

Parameter	Value
P_t	30 dBm (1 W)
P_N	-100 dBm (10^{-10} mW)
r	50 (m)
$\bar{\theta}$	60
r_{LoS}	250 (m)
ρ	0.0156 (per sq. m)
β	1.5625e-04 (per sq. m)
λ	1 cm (30 Ghz)

Table 2.2: Parameters used for SimRIS simulation plots

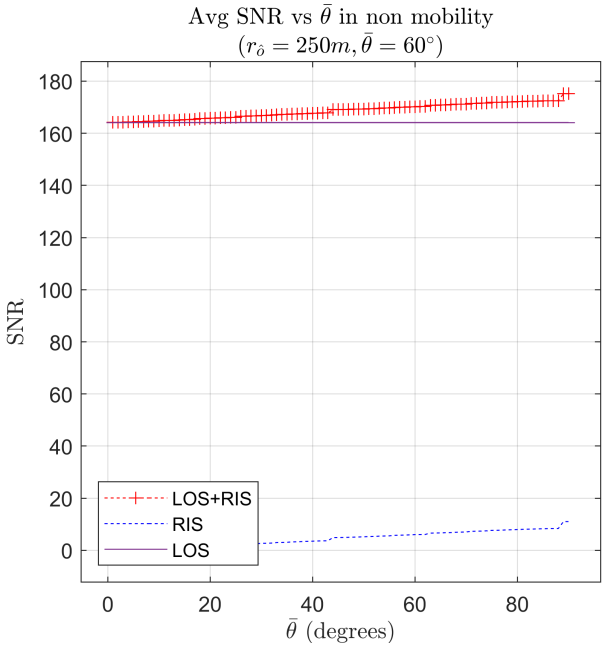


Figure 2.5: S_r Vs $\bar{\theta}$ for $\alpha = 2$

Observations:

1. We can clearly observe that the contribution of the power received through the RIS panels is negligible in comparison to the power received from the LoS path for both plots.

Based on our observations of the plots of S_r Vs r and $\bar{\theta}$ in both the theoretical and SimRIS simulation, we can conclude that in a scenario with free space path loss in the LoS path, the use of

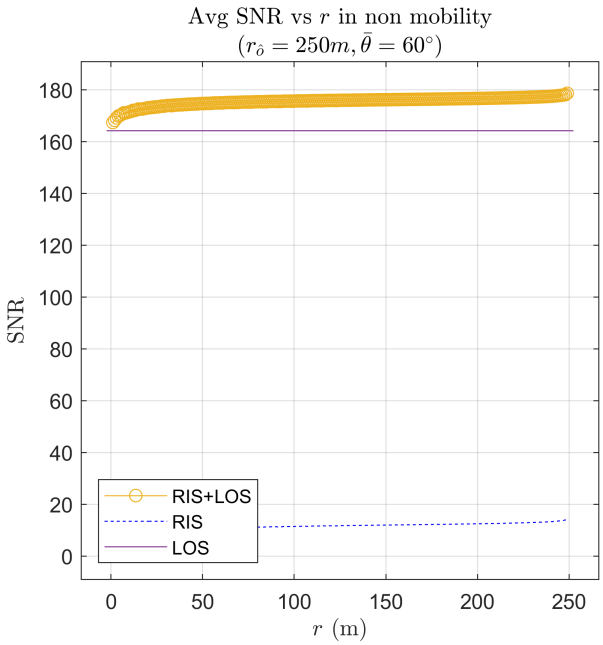
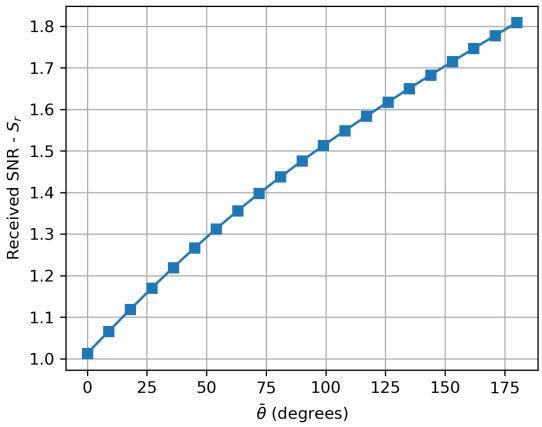


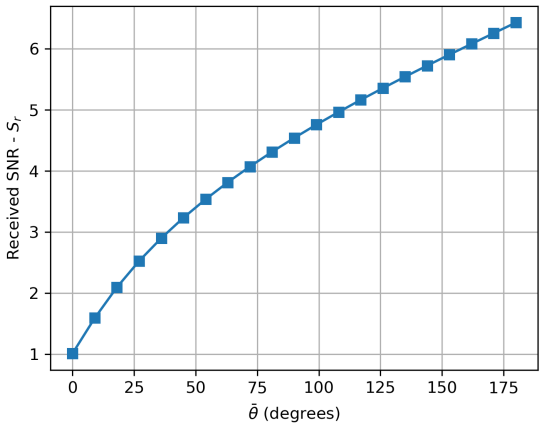
Figure 2.6: S_r Vs r plot for $\alpha = 2$

RIS panels would not be successful in increasing the SNR and improving the communication.

2.5 Analysis assuming $\alpha = 4$ for direct UAV-BS link



(a) S_r Vs $\bar{\theta}$ for $r = 20$m, $r_{LoS} = 50$m, $\rho = 0.1$ per sq. m



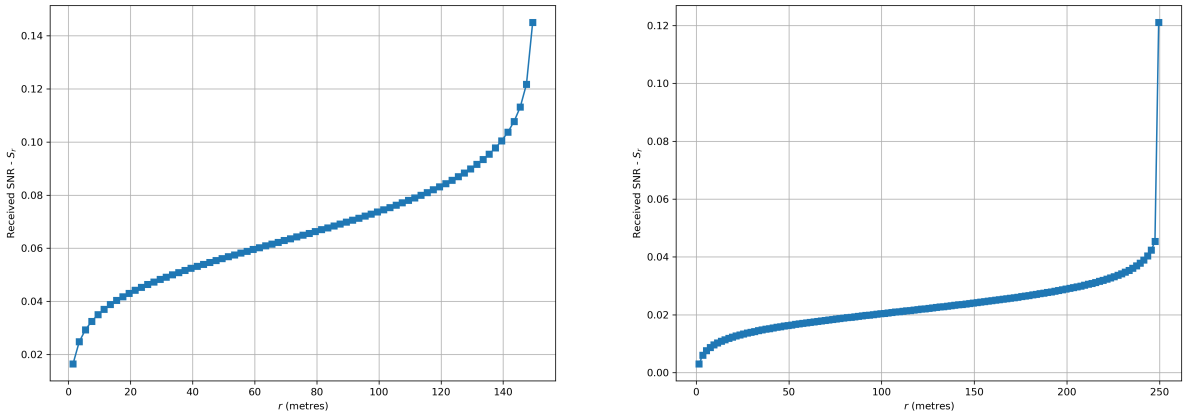
(b) S_r Vs $\bar{\theta}$ for $r = 40$m, $r_{LoS} = 50$m, $\rho = 0.5$ per sq. m

Figure 2.7: Received SNR S_r Vs capture angle $\bar{\theta}$ plot for $\alpha = 4$

Following the conclusion from the previous section, we analysed the system with LoS path loss exponent $\alpha = 4$. This means we assumed that the LoS path does not follow free path loss model and rather has a number of scatterers in the path. We can repeat the received SNR analysis as the previous section to understand the benefit of using RIS panels in the new system.

The theoretical plots for received SNR, S_r, versus the variation in capture angle $\bar{\theta}$ is shown in figure 2.7. Unlike the previous section, the power through LoS does not dominate the received SNR and we can observe that at $\bar{\theta} = 180^\circ$ the improvement over no-RIS scenario ($\bar{\theta} = 0^\circ$) is around 80% when $r = 20$m and around 800% when $r = 40$m.

Similarly, the theoretical plots for received SNR, S_r, versus the variation in capture radius r is shown in figure 2.8. We can observe that the improvement using RIS over the no-RIS scenario is significant.



(a) S_r Vs r for $\bar{\theta} = 60^\circ$, $r_{LoS} = 150$ m, $\rho = 0.1$ per sq. m

(b) S_r Vs r for $\bar{\theta} = 60^\circ$, $r_{LoS} = 250$ m, $\rho = 0.1$ per sq. m

Figure 2.8: Received SNR S_r Vs capture radius r plot for $\alpha = 4$

Observing the SimRIS simulation plots for S_r Vs $\bar{\theta}$ and S_r Vs r as shown in figures 2.10 we can see that the the power through the RIS path contributes the maximum (almost entirely) to the total received SNR. This strengthens our point that the use of RIS panel to assist communications will be helpful in a feasible way (in the sense of number of RIS panel deployed for a UAV-BS pair) in cases where the LoS path consists of scatterers and do not dominate the power through RIS path by orders of magnitude, ex - our setup with $= 4$.

Observations from S_r Vs $\bar{\theta}$ plot:

1. The variation in received SNR is near linear with change in $\bar{\theta}$ from 0° to 180°.

Theorem 2.5.1 *For system A with capture angle $\bar{\theta}_A$, there exist a ρ_A such that the performance of system A in terms of received SNR is equal to the performance of system B with parameters $\bar{\theta}_B$ and ρ_B.*

Proof 2.5.1 *See appendix A.*

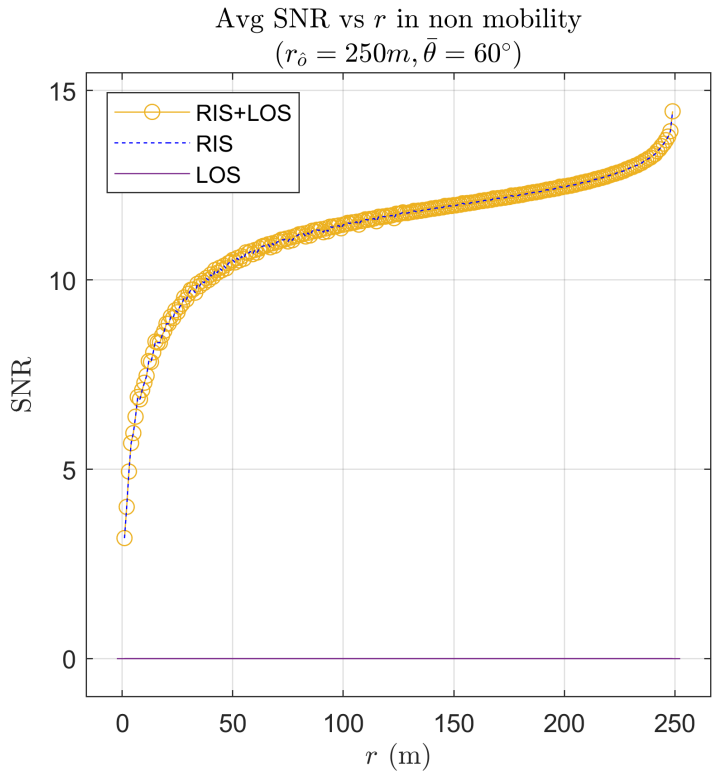


Figure 2.9: S_r Vs r plot for $\alpha = 4$

Based on the observations from the figures 2.8 and 2.9, we can observe a characteristic nature of the curve for variation of S_r with r. The curve seems to have two knee points because of which we named its analysis as **knee point analysis**.

2.5.1 Knee Point Analysis

We can observe the presence of two kinds of regions in these plots: slowly varying linear region in the middle and quickly varying non-linear region towards the end-points. The non-linear regions are the one of interest since they provide more advantage with a lesser increase in r. We will refer the linear to non-linear transition points as *knee points*. The knee point can be defined as the point where the slope is $x\%$ greater than the lowest value of the slope of the S_r Vs r curve. These points can help the controller decide the optimal system operation point by allocating the RIS through tuning $(r, \bar{\theta})$ for different BSs. The nature of these curves can be explained by looking at the rate of change of average SNR as a function of r.

By differentiating (2.6), we have,

$$\frac{\partial S_r}{\partial r} = \left(\frac{\lambda}{4\pi}\right)^4 \frac{P_t N \rho}{P_N r_{LoS} r^2} \frac{4\tan^{-1}\left(\frac{(\delta+1)\tan\left(\frac{\bar{\theta}}{2}\right)}{\sqrt{\delta^2-1}}\right)}{\sqrt{\delta^2-1}} \tag{2.9}$$

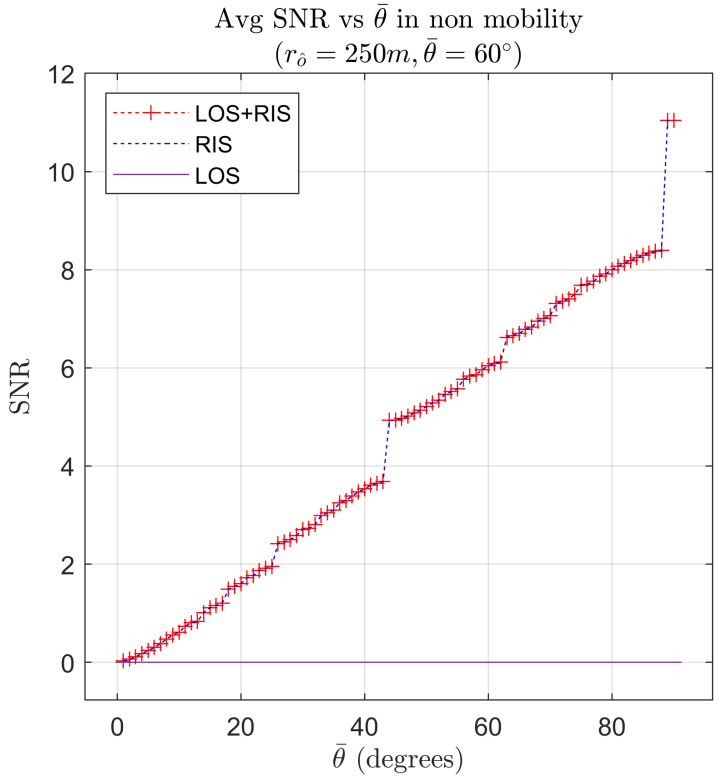


Figure 2.10: S_r Vs $\bar{\theta}$ for $\alpha = 4$

where $\delta = \frac{(r_{LoS}^2 + r^2)}{2 r_{LoS} r}$.

The slope plot using equation 2.9 is shown in figure 2.11.

Key points:

1. The rate of change of received power is very high nearer to the location of BS and UAV, see Fig. 2.11. This suggests to arrange RIS elements near and around these locations and not at other places as the return due to complete spatial distribution is not proportionally high.

2. The two knee points do not show symmetrical behaviour as seen from the UAV's and the BS's location. The rate of change or slope of the curve as in Fig 2.11 approves the un-symmetrical behaviour. The rate of change is more around the BS and the UAV but the greater of the two varies with $\bar{\theta}$ as shown in Fig 2.11.

3. The knee point can serve as a control parameter in system design for RIS distribution and allocation to meet the desired increment in SNR by RIS addition.

4. The existence of the knee points is not dependent on the location of the UAV and BS. The shape of the curve will exist until the near field assumption in Friss equation is not violated.

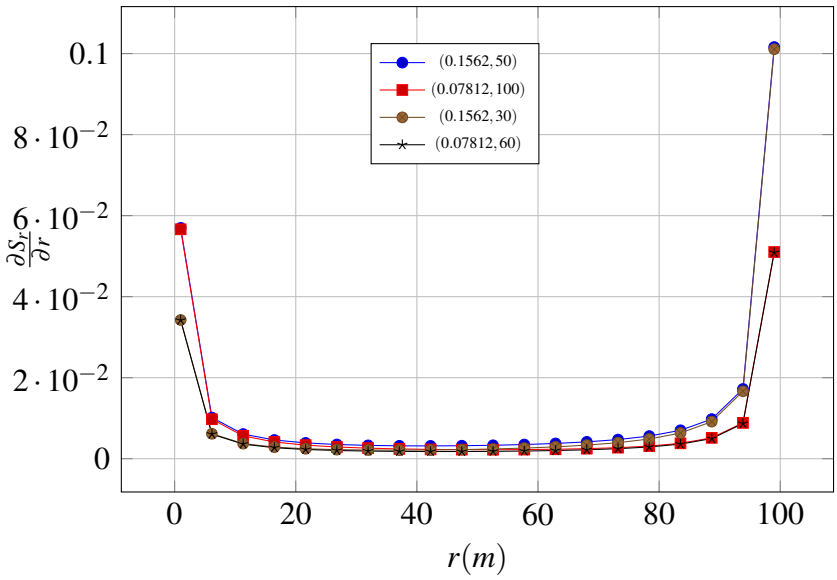


Figure 2.11: Rate of change of SNR, S_r, vs r for different value of $(\rho, \bar{\theta})$ with r_{LoS} = 100m

2.6 Mobility Scenario

The previous section considered the UAV to be stationary and the analysis was based on this assumption. In general, a UAV will move around and connect to a number of BSs as the nearest BS changes over flight. This section deals with such scenario where the UAV is mobile. In such a case, the average SNR received will vary randomly, depending on the instantaneous BS-UAV distance. We assume that the UAV follows a 2-dimensional Brownian motion i.e. random motion in X-Y plane. The BS are distributed as PPP with density β. Using the property of PPP, the average SNR is given by equation 2.10.

$$S_{av}(\rho, r, \bar{\theta}) = \int_{r}^{\infty} S(r_{LoS}, \rho, r, \bar{\theta}) f(r_{LoS}) dr_{LoS}, \tag{2.10}$$

where the expectation is over the random variable r_{LoS}, for which the pdf is given by,

$$f(r_{LoS}) = \frac{2\pi\beta r_{LoS} e^{-\beta\pi r_{LoS}^2}}{e^{-\beta\pi r^2}}; \quad r < r_{LoS} < \infty, \tag{2.11}$$

The denominator accounts for the conditioning of $r < r_{LoS} < \infty$ which means that the BS cannot lie inside the $(r, 2\bar{\theta})$ sector. The variation of average SNR vs β are plotted in Fig. 2.12. We can see that the S_{av} improves with the increase in density of the transmitters (β), for both RIS and no-RIS cases. This can be explained as when β increases, the BSs are spread in a more dense manner. This results in decrease in the UAV-BS distance and therefore, increase

in the received power and SNR. The relative benefit of RIS over no-RIS scenario can also be observed in Fig. 2.12, , for a fixed β, average received SNR increases with ρ. Fig. 2.13 shows the simulations results using SimRIS for validation. Amongst the three signal shown in Fig 2.13, the LoS+RIS, aggregate signal, is around 2x of the LoS contribution, which suggests powerful improvement by RIS.

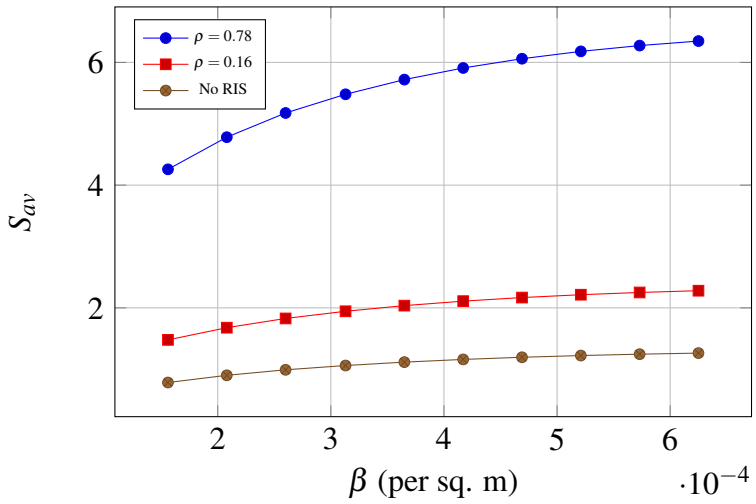


Figure 2.12: Avg SNR with mobility, S_{av}, vs β for various configurations of ρ when $(r, \bar{\theta})$ is $(40, \pi/3)$.

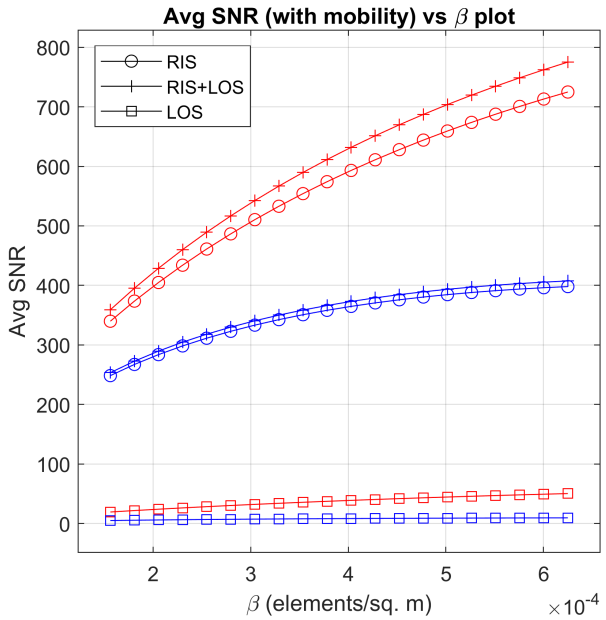


Figure 2.13: SimRIS plot of avg SNR with mobility, S_{av}, vs β with the red and the blue curves representing signals for $r = 10$m and $r = 20$m respectively.

CHAPTER 3

Introduction - Multi-AP Coordination

The idea to choose the field of multiple access point transmission as part of my thesis originated out of the the discussions around standardisation process of IEEE 802.11be. The newest member of the IEEE 802.11 family aims to achieve extremely high throughput, around 40Gbps which is equivalent to the speed of Thunderbolt 3. To achieve this researchers have to explore newer paradigms one of which is to coordinate several access points and use it to serve one single user. Our institute, IIT Bhilai, has campus-wide WiFi coverage using dozens of WiFi access points all over the place. Being able to see and relate to the use case everydy, the idea of research in the domain was exciting and thought provoking.

Since its introduction in the 1990s, WLAN has steadily increased its market share and is becoming increasingly crucial for offering wireless data services via Wi-Fi technology. Wi-Fi technology is becoming increasingly important as the primary access network for homes, businesses, and hotspots. According to a recent Wi-Fi Alliance survey, there are already more than 9 billion Wi-Fi devices in use globally, including personal computers, cellphones, TVs, tablets, sensors, and so on.

Over several years, wireless networking technologies have seen tremendous efforts of technological improvements to ensure seem-less sharing of information. Mobile communication technologies have come a long ways starting from AMPS to 5G and talks around 6G as of today covering a wide range of use cases as industry and demands have evolved. On the same lines, WiFi has also improved massively, starting from 802.11-1997 all the way to the latest centre of research attraction - 802.11be. In these years, a lot of new ways to use the existing technologies and new technologies have been tested and used but the idea of coordinating several transmitters to send data to a single user did not get much traction. With the wireless spectrum cluttered with huge loads of traffic, the need to work on spectral efficiency is more important than ever. In a general 2-base station and user setup, there are places far away from both the base stations and hence suffer from poor signal to interference and noise ratio (SINR). These spots are called the cell edges as they represent the edge of the cell (region of service) of a base station. Multi-AP (multiple access point) coordination helps achieve better spectral efficiency to users located at the cell edge.

3.1 Background

3.1.1 WiFi 7

The 802.11be project has set lofty goals for faster nominal data speeds, more spectrum efficiency, improved interference mitigation, and RTA support. To achieve these goals, the 802.11 Working Group has reviewed around 500 suggestions from many fields, each of which may be linked to one of the seven fundamental Wi-Fi 7 advancements. Some of the main features are pictorially represented in figure 3.1 are:

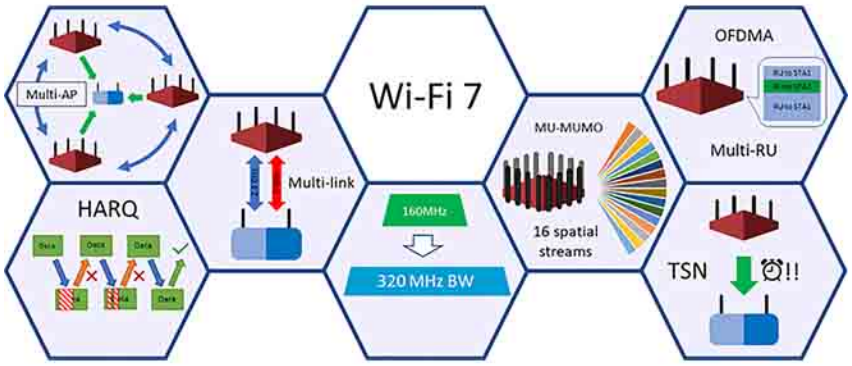


Figure 3.1: Illustration of main features of WiFi 7 [7]

1. **EHT PHY**: Wi-Fi 7 has been certified to scale the PHY of earlier Wi-Fi standards by doubling both the bandwidth and the number of SSs in MU-MIMO, resulting in a notional throughput increase of 22=4 times. PHY also adds higher-rate MCSs by leveraging 4K-QAM, which increases nominal throughput by 20

2. **Enhanced OFDMA**: Wi-Fi 7 addresses the issues related to insufficient flexibility of OFDMA, which was introduced in 802.11ax.

3. **Multi-Link Operation**: Wi-Fi 7 strives to find such a level of synchronization between the links that allows efficient use of the channel resources and does not suffer from interference in dense deployments.

4. **Multi-AP Cooperation**: Multi-AP collaboration is another significant breakthrough introduced in 11be. By this point, the 802.11 Working Group had mostly focused on fully-distributed coordination between neighbouring APs. This requires significantly closer coordination among neighbouring APs, including coordinated scheduling, beamforming, and even distributed MIMO systems.

Figure 3.2 lists major innovations of IEEE 802.11be and the features that are proposed to implement them.

Innovation \ Target	Nominal Throughput	Interference Mitigation	Spectrum Effeciency	Real-Time Applications
EHT PHY	4096 QAM, 320 MHz, 16x16 MU-MIMO		EHT Preamble	
EDCA with 802 TSN Features				IEEE 802 TSN, Faster Backoff, New Access Categories, TXOP capturing
Enhanced OFDMA		Preamble puncturing	Multi-RU, Direct links	Enhanced UORA
Multi-link Operation	Multi-link Architecture	Synchronous Channel Access	Virtual BSS	Asynchronous Channel Access, Packet Duplication, Queue Management, Dynamic Link Switching
Channel Sounding Optimization			Implicit Sounding, Explicit Feadback, Channel Estimation	
Advanced PHY	Full Duplex		HARQ, NOMA / SOMA	
Multi-AP Cooperation		Null steering, Co-OFDMA, CSR	Distributed MU-MIMO, Multi-AP Sounding	Joint Reception

Figure 3.2: Major innovations and features to implement them in Wi-Fi 7 [7]

3.1.2 Coordinated transmission

As mentioned in the last point of the list of some of the main features of Wi-Fi 7, coordinated transmission is a research topic with lots of active interest. The idea of coordinating different access points (APs) to transmit in a way that a single user gets the same signal after phase and beamforming adjustments over Wi-Fi, is similar to the idea of CoMP [1] used in LTE technology. The authors in paper [5] investigate the Joint Transmission (JT) CoMP performance in a C-RAN implementation of LTE-A HetNet with large CoMP cluster sizes. The paper [9] deals with important specification support to realize the benefits of cooperative transmission in the downlink and cooperative reception in the uplink for 3GPP Release 11.

3.1.3 Use cases

The use cases of coordinated transmission can be many, but the use case that we want to focus on in the thesis is to support QoS requirements for the cell-edge users. In a classical cell type communication setup, where the cell denotes the range around the transmitter where the signal can be received successfully, the users at the cell edge are the ones who do not get enough power from any of the transmitters as all the nearby transmitters are far enough for the received SNR to fall below the requirements. By using coordinated transmission, we use multiple transmitters to send the same signal in such a way that it is added constructively at the receiver's end. The power of the received signal is increased which increases the received SNR and corresponding data rate.

3.2 Organisation of Thesis for Multi-AP coordination

The part of the thesis explaining the research on Multi-AP coordination is divided in 4 chapters namely - Introduction, Analysis of Multi-AP Coordination in 1-D Corridor, Fluid Model - JT Scenario and Fluid Model for JT+SAPI Scenario. The introduction gives the idea, motivation, related work and use cases for the concept of multi-AP coordination. Chapter 2 consists of a description of the system model with definition and detailed pictorial representations of the different modes of service. It also consists the capacity calculation formulas in general and with Modulation Coding Scheme, and analysis on the region that can be benifitted by using Joint Transmission (JT). Chapter 3 consists on the details of a scheduling scheme for serving a 2-STA 2-AP system in JT mode. It describes the queue model of the system and conditions for stable and stationary queues are found and verified with simulations. Chapter 4 extends the ideas proposed in Chapter 3 by describing the a general scheduling scheme for the same system with STAs served in both joint and uncoordinated transmissions. Queue model is analysed for stability and conditions such that the queue length remain stationary are found out. The analysis on delay and jitter is also explained using simulations.

CHAPTER 4

Analysis of multi-AP coordination in 1-D corridor

In this chapter, we have proposed a system model setup in a 1-dimensional cooridor in SISO (single input single output) mode to analyse the different modes consisting of uncoordinated and coordinated transmission. We will compare these on the basis of capacities achieved from the perspective of both the user and the system.

4.1 Modes of service

We assume three different modes of transmission from the AP to the STA in uncoordinated and coordinated approaches as suggested in [12]. In the uncoordinated approach, the APs in the system is only responsible for signalling and transmission to a STA connected to it. It's functioning does not get affected by the working of the other APs in the system. On the other hand, the coordinated approach involves additional signalling and synchronisation for 2 or more APs to transmit the same signal to a single STA such that the these signals from the APs add constructively at the STA increasing SNR and capacity.
The methods under uncoordinated approach are:

1. **Single AP Transmission (SAPT)**: In this mode, one AP in the system serves only one STA on a channel where no other AP is transmitting, making the system noise-limited. ex: AP1 transmitting STA1 on channel 1 and AP2 transmitting STA2 on channel 2. The respective transmissions on separate channels causes no interference for the 2 STAs.
2. **Single AP transmission with Interference (SAPI)**: In this mode, two or more APs transmit to serve one STA each on the same channel causing the STAs served in this mode to experience interference. This is a interference-limited scenario. ex: AP1 and AP2 serve STA1 and STA2 respectively on the same channel.

The method under coordinated approach is called as *Joint Transmission* is explained as:

1. **Joint Transmission (JT)**: In this mode, two or more APs can serve a single STA by

transmitting the same message signal in such a way that the signals from different APs add constructively at the STA.

4.2 Pictorial understanding of different modes of service

We can understand the working details of each of the different modes or service in uncoordinated and coordinated fashion, using an example. Let us consider a 2-AP, 2-STA and 2 channel system. Figure 4.1 shows the 2-APs serving the 2-STAs in SAPT mode. The oval around the AP represents the region where the signal transmitted by the AP is detectable. Different colors in the two ovals represents the transmission of signal on two non-interfering channels and the intensity of the color fades away from the AP to represent large scale fading in the transmitted signal. The dashed arrows represent the direction of the intended receiver for the signal i.e. for downlink STA1 is the intended reciever for AP1 and vice-versa for uplink. Different colors of the dashed line represent that the messages being transmitted are for different intended users. While same color of the dashed line means the same message is being transmitted for the same intended user.

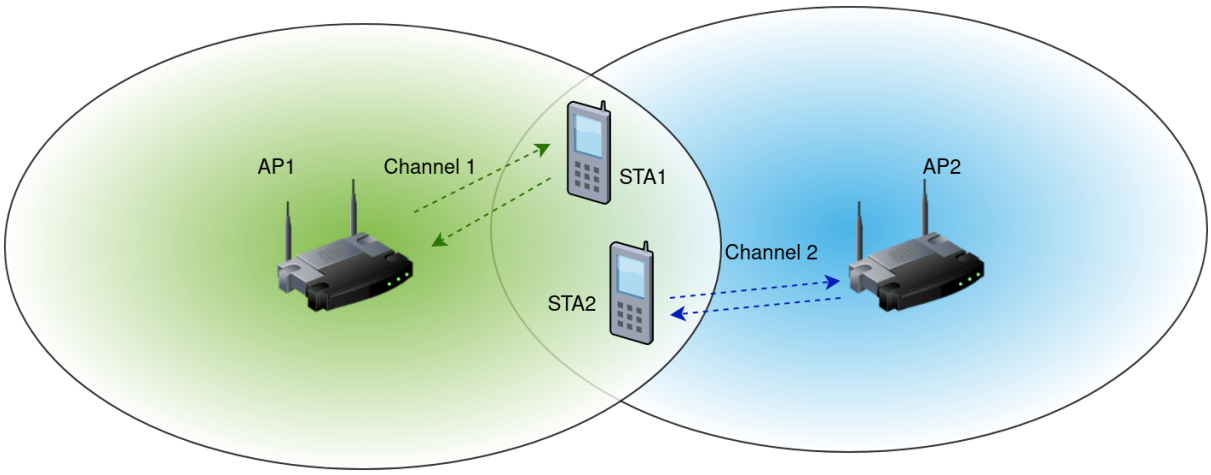


Figure 4.1: Example showing working of SAPT

From the figure and using the color scheme explained, we can observe that each of the AP is transmitting on a non-interfering channel due to which even though STA2 will receive the signal from AP1 intended for STA1 and similarly STA1 will receive the signal from AP2 intended for STA2, both of these STA will not face any interference.

From figure 4.2, the orange color of the oval representing the range of the APs indicates the the APs are using the same channel for transmission. The green and blue dashed arrow represents that the 2 APs intend to communicate to 2 different STAs. This represents service in SAPI mode as each of STA1 and STA2 will suffer from interference due to intended signals for the other STA.

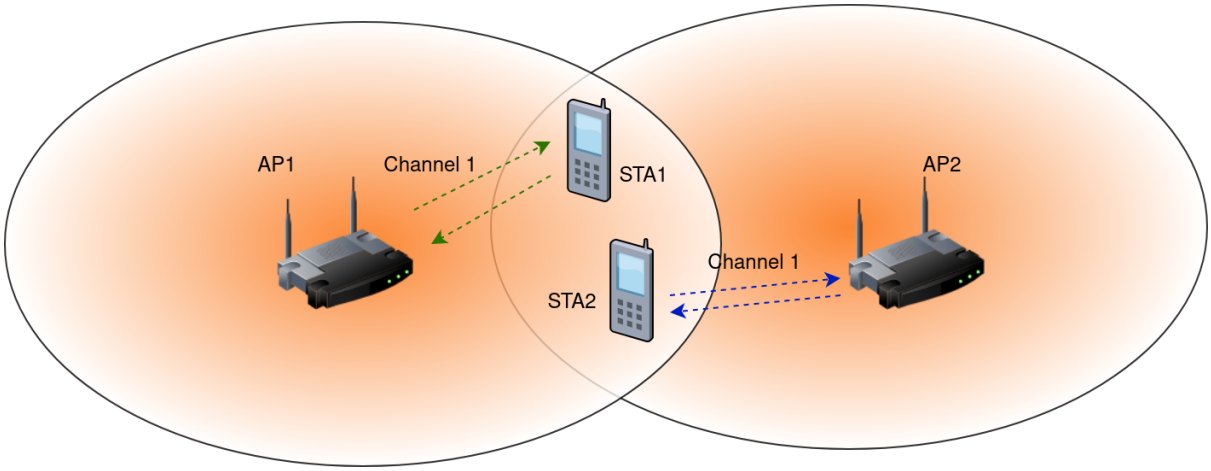


Figure 4.2: Example showing working of SAPI

We can observe from figure 4.3 that the color of the oval is the same for both APs representing that they are using the same channel for transmission. The same colored dashed arrow represents that the same message is being transmitted for the same intended user, STA1 in this case. This represents an example for service in JT mode as 2 APs coordinate to serve a single user.

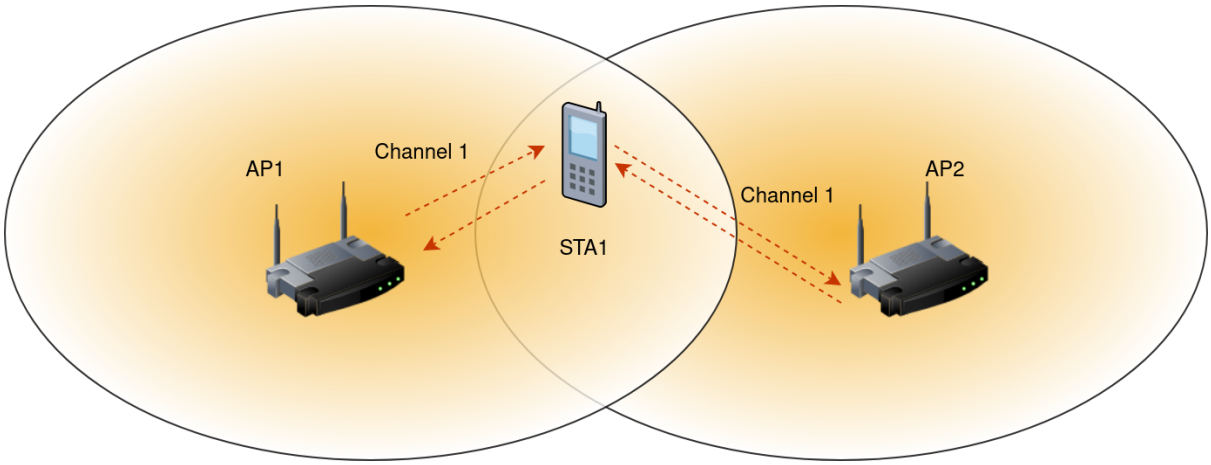


Figure 4.3: Example showing working of JT

4.3 System Model

The system model consists of a 1-D corridor of STAs separated by d metres each ($d = 1$ for our analysis) with two APs, one placed either side of the corridor at a distance 1m from the nearest STAs, as shown in figure 4.4. Each AP will have its own scheduling. A **complete round of service** is defined to be achieved when all STAs have been served once. We assume a **free space path loss model** for signal propagation in the system

Defining notations for a general system, lets consider N_{sta} number of STAs in the system and the set of all the STAs be denoted by $S = \{STA_i\}, i\varepsilon\{1,2,...,N_{sta}\}$. Let us denote the set of

all the APs in the system as $A = \{AP_i\}, i\varepsilon\{1,2,...,N_{ap}\}$ where N_{ap} is the total number of APs ($N_{ap} = 2$ in our case). Let P_t be a set containing the transmit powers of all the APs with $P_t[i]$ denoting the power for AP_i. The distance between AP indexed i and STA indexed j is denoted by $d_{i,j}$. Let N_{ch} denote the number of non-overlapping channels allocated for the system.

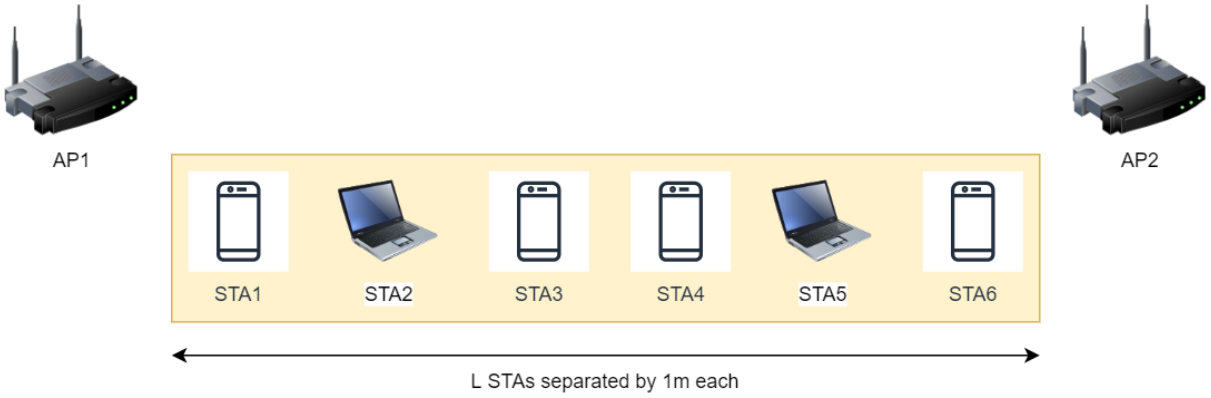


Figure 4.4: Illustration of the 1-D corridor of STAs

Since there are only two APs in the system a maximum of two channels ($N_{ch} = 2$) will be required. Based on the number of channels available, there can be two scenarios:

1. **1-channel**: there are 2 APs and 1 channel. They have to serve the given STAs only in modes SAPI or JT. Such a system will be called **JT-enabled** system but do not confuse it as all STAs will be served in JT.

2. **2-channel**: there are 2 APs and 2 channels, i.e. both can serve an STA on a separate channel using only SAPT. Such a system will be called **JT-disabled** system.

4.4 Channel Capacity

Channel capacity is a measure of how good a channel is in terms of the amount of information we can transmit through the channel such that the signal is decodable at the receiver's end (using expressed in bits per second). The channel capacity is calculated using the **Shannon-Hartley's theorem** which gives us the tightest upper bound on rate with which information can be communicated with an arbitrarily small error rate, in terms of signal to noise ratio (SNR) and bandwidth used. Assuming P_r, N_0 and B represents the received signal power, noise power and the bandwidth used respectively, the equation for capacity is given by:

$$C = B.log_2(1 + \frac{P_r}{N_0}) \tag{4.1}$$

4.4.1 Capacity for different modes of service

The model assumes a free space path loss model where the line of sight signal is the only signal considered for analysis. We can use free space path loss model ($\alpha = 2$ in equation 2.1) to analyse the received power and find the corresponding SNR/SINR values. The received power when an AP_i transmits to a STA_j is given by:

$$P^r_{i,j} = D_t D_r \left(\frac{\lambda}{4\pi d_{i,j}} \right)^2 P^t_i \tag{4.2}$$

,where D_t, D_r are the isotropic directivities of transmitter and receiver respectively. In our case, we assume them to be 1. Using Equation 4.2, the SNR equation for SAPT for an STA indexed j is:

$$\gamma^{SAPT}_{i,j} = \frac{P^r_{i,j}}{\sigma^2} \tag{4.3}$$

where σ^2 is the power of the noise. The SINR equation for STA indexed j served in SAPI mode is given by:

$$\gamma^{SAPI}_{i,j} = \frac{P^r_{i,j}}{\sigma^2 + P^r_{k,j}}, k \neq i. \tag{4.4}$$

The equation of joint transmission (JT) can be written as:

$$\gamma^{JT}_{j} = \frac{P^r_{1,j} + P^r_{2,j}}{\sigma^2} \tag{4.5}$$

The capacity equations can be derived using the shannon capacity formula given by equation 4.1 with bandwidth B as (subscripts are omitted for generality):

$$C^{SAPT} = B.log_2\left(1 + \gamma^{SAPT}\right) \tag{4.6}$$

$$C^{SAPI} = B.log_2\left(1 + \gamma^{SAPI}\right) \tag{4.7}$$

$$C^{JT} = B.log_2\left(1 + \gamma^{JT}\right) \tag{4.8}$$

MCS index	Modulation	Code rate	N_{DBPS}	Spectral efficiency (bps/Hz)	Required SNR (dB)
0	BPSK	1/2	26	0.5	−3.83
1	QPSK	1/2	52	1	0
2	QPSK	3/4	78	1.5	2.62
3	16-QAM	1/2	104	2	4.77
4	16-QAM	3/4	156	3	8.45
5	64-QAM	2/3	208	4	11.67
6	64-QAM	3/4	234	4.5	13.35
7	64-QAM	5/6	260	5	14.91
8	256-QAM	3/4	312	6	17.99
9	256-QAM	5/6		Not valid	

Figure 4.5: MCS table for reference

4.4.2 Allocation to APs

The different STAs have to be allocated to the 2 APs available in order to be served in SAPT, SAPI or JT. For a STA at index i we can represent the capacities achieved when served by AP1 in SAPT mode as $C^{SAPT}(d_{1,i}$. For an STA served by AP1 interfered by AP2 in SAPI can be denoted as $C^{SAPI}(d_{1,i},d_{2,i})$ and capacities achieved in JT mode as it is served by both AP1 and AP2 as $C^{JT}(d_{1,i},d_{2,i})$ respectively. The capacities if the STA is served by AP2 can be found similarly. Let the best capacities of STA indexed i for SAPT, SAPI and JT modes be denoted by C_i^{SAPT}, C_i^{SAPI} and C_i^{JT} respectively, is given by equation 4.9,4.10 and 4.11.

$$C_i^{SAPT} = max(C^{SAPT}(d_{1,i}),C^{SAPT}(d_{2,i}) \tag{4.9}$$

$$C_i^{SAPI} = max(C^{SAPI}(d_{1,i},d_{2,i}),C^{SAPI}(d_{2,i},d_{1,i})) \tag{4.10}$$

$$C_i^{JT} = C^{JT}(d_{1,i},d_{2,i}) \tag{4.11}$$

By using the above equations, we can know that if the STA is served in a particular mode, which AP(s) will it be allocated to for service.

4.4.3 Capacity with MCS

It is to be noted that the capacity given by the formula is only the upper limit and the way to achieve it is not specified. It means that the practical capacities should be analysed using some modulation scheme. Let $M = \{M_i\}$, $i\varepsilon\{1,2,...,N_{mcs}\}$, be the set of all modulation and coding rate pair allowed in the system, where N_{mcs} is the total number of such pairs. MCS (modulation coding scheme), minimum required SNR and corresponding capacities practically achieved using the particular MCS is shown in Figure 4.5 for 802.11ax.

A simple algorithm to find out the best achievable capacity using a particular modulation coding pair from set M is proposed next. The algorithm takes the following inputs:

1. Capacity in SAPT, SAPI and JT for all STA.
2. The MCS table including the minimum SNR/SINR with corresponding spectral efficiencies.

Let us denote the set of best capacities for all the STAs obtained using equations 4.9, 4.10 and 4.11 as $C^{SAPT} = \{C_i^{SAPT}\}$, $C^{SAPI} = \{C_i^{SAPI}\}$ and $C^{JT} = \{C_i^{JT}\}$, $i\varepsilon\{1,2,...,N_{sta}\}$ for SAPT, SAPI and JT modes respectively.

4.5 Analysis of 1-channel case

Let us assume only one channel in the system i.e. $N_{ch} = 1$, with SAPI and JT the only two modes of service possible. We can analyse the two modes on the basis of **system capacity**.

Definition 4.5.1 ***System capacity****: the sum of capacities delivered using both the APs at a given instant. Let us denote system capacities in JT and SAPI modes as $C_{sys}^{JT}, C_{sys}^{SAPI}$ respectively. These are given by equation 4.14 and 4.13.*

$$C_{sys}^{SAPT} = 2 * C_i^{SAPT} \tag{4.12}$$
$$C_{sys}^{SAPI} = 2 * C_i^{SAPI} \tag{4.13}$$
$$C_{sys}^{JT} = C_i^{JT} \tag{4.14}$$

It In case of JT, the system capacity is same as the capacity of the STA as both the APs are busy serving one STA. Whereas, in case of SAPI, the system capacity will be double the capacity served by 1 AP to 1 STA since the second AP will serve another STA placed symmetrically (the double concept comes from symmetricity).

$$Mode = \begin{cases} JT, & C_{sys}^{JT} > 2C_{sys}^{SAPI} \\ SAPI, & C_{sys}^{JT} <= 2C_{sys}^{SAPI} \end{cases} \tag{4.15}$$

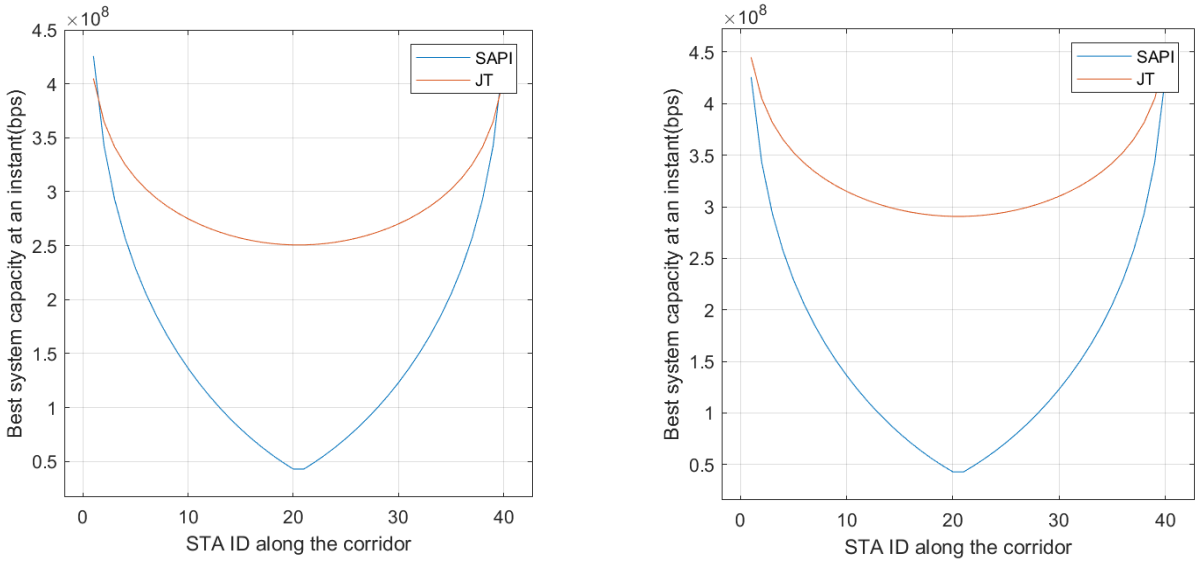
The system capacity for a 1-d corridor with $N_{sta} = 40$ for transmit power of 5 mW and 20 mW is plotted in figure 4.6a and 4.6b respectively. **Observations:**

1. There exist a region defined by two crossover points where system capacity in JT exceeds that in SAPI, i.e. $C_i^{JT} > 2 * C_i^{SAPI}$.

2. The region where JT performs better than SAPI in terms of system capacity, increases as we increased transmit power from 5mW to 20mW.

3. The figure does not include concepts of time durations (slots) of service, fairness or anything else. But it gives us a decision that **whenever we will serve STA i, we will serve it according to the mode which is better according to the graph**.

4. The scheduling decisions based of fairness or some other constraint can be introduced using other parameters, but the decision of mode is inferred from the graph.



(a) Power used - (5,5) mW

(b) Power used - (20,20) mW

Figure 4.6: System capacity for varying values of transmit power in bits per second)

The best achievable capacities in different modes for all the STAs is shown in Figure 4.7. We can see that the smooth curves are no longer smooth if we talk about practically achievable capacities.

4.6 JT region

Let us denote the set of STAs in the corridor served in JT mode as S_{JT} and the number of such STAs as N_{sta}^{JT}. $S_{JT} = \{i : C_i^{JT} > 2 * C_i^{SAPI}\}$. We want to analyse the variation of N_{sta}^{JT} as N_{sta} increases and the behavior of the change of the ratio $r = \frac{N_{sta}^{JT}}{N_{sta}}$. The plots for N_{sta}^{JT} Vs N_s is shown in figure 4.8 and the plots for r Vs N_{sta} is shown in figure 4.9. Two scenarios of crossover comparison is plotted:

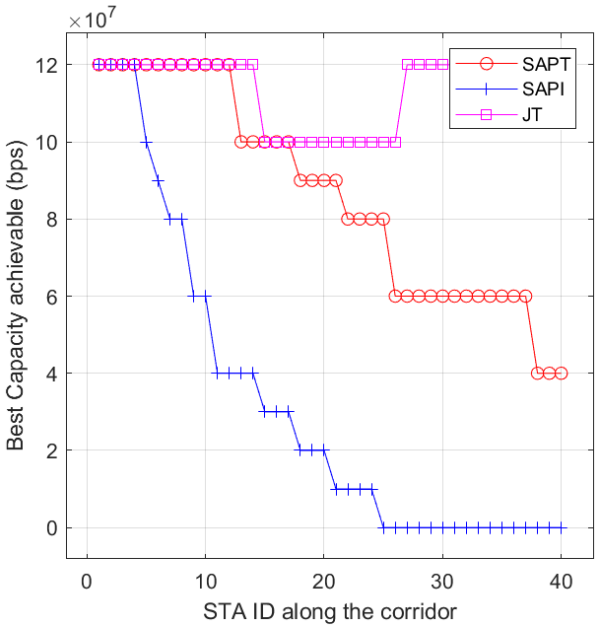


Figure 4.7: Best capacities for all the STA locations taking $N_s = 40$ taking MCS with a noise floor of $-70dBm$.

1. **Raw values**: the values of the capacity for mode selection, as in equation 4.15, is done using shannon's capacities.

2. **With MCS**: the values of capacity for mode selection uses the best capacities achieved at an STA location for different modes. This gives a more **realistic** analysis in terms of what is achievable in an actual transmission configuration.

Remark 4.6.1 ***Noise power****: the noise power (σ^2) consists of two components namely thermal noise and receiver's noise. The thermal noise value for BW = 20Mhz is taken and the receiver's noise is taken as 6dBm. This makes* $\sigma^2 = -94dBm$.

The observations from these plots are:

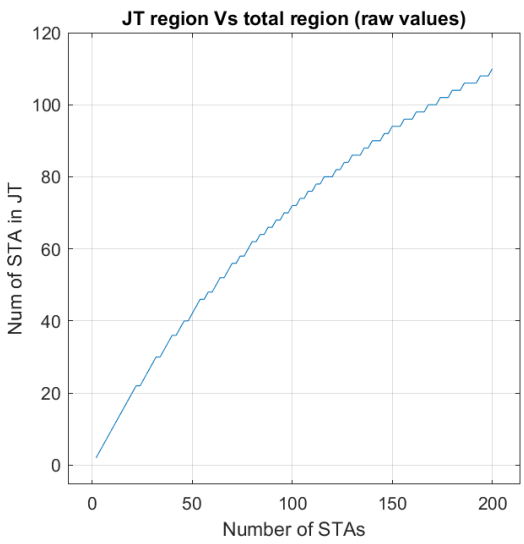
1. The figure 4.8(b) shows that the rate of change $\frac{d(N_{sta}^{JT})}{d(N_{sta})}$ is constant and this constant depends on the transmit power and σ^2. (we do not have a proof but we think so)

2. This rate of for raw values is near linear as in with MCS case. It decreases as N_s increases.

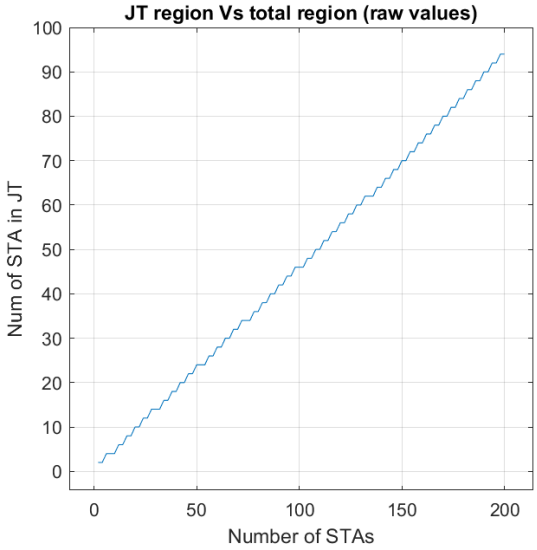
3. N_{sta}^{JT} is more than $0.5N_{sta}$ in the raw values case and around $0.45N_{sta}$ in with MCS case.

4. Figure 4.9 supports the claim that more than 55% of the total region should be served by JT, where the 55% value corresponds to $N_{sta} = 200$. This is a lower bound as a 1-D corridor of 200m length is unrealistic and for practical situations, it will be maximum around 30-35 m. The same is true for with-mcs case and r lies around 0.45.

5. The figures supports the claim that for realistic values of noise, power and length of corridor, using the concept of JT will help improve the capacity of **individual STAs**.

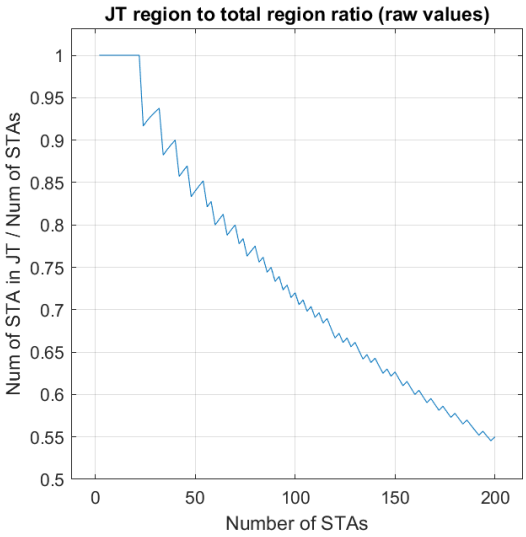


(a) N_{sta}^{JT} with raw values

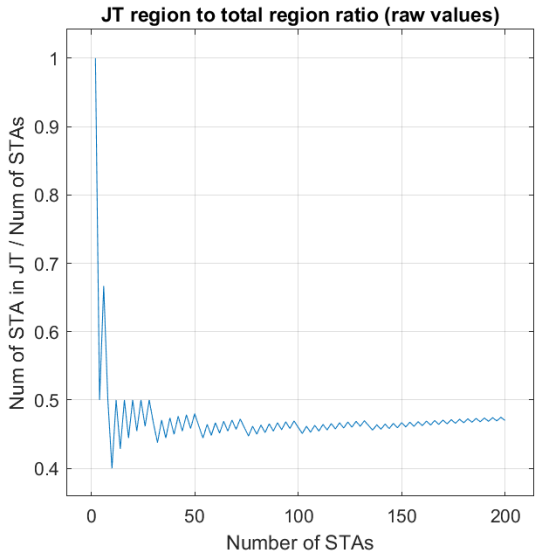


(b) N_{sta}^{JT} with MCS

Figure 4.8: N_{sta}^{JT} Vs N_{sta}



(a) r with raw values



(b) r with MCS

Figure 4.9: r Vs N_{sta}

CHAPTER 5

Fluid model - JT scenario

Once we have the theorya to obtain the SNR/SINR and corresponding capacities for the different modes i.e. SAPT, SAPI and JT, we can focus on the way the STAs are served and scheduled. Assuming the allocation done while finding the best capacity achieved using equations 4.9, 4.10 and 4.11, we can define a scheduling scheme which we name **fluid model**.

5.1 Queue model

Each STA has a corresponding data queue maintained by the APs from which the data packets are served to the particular STA. Let the set of queues be represented by $Q = \{Q_i\}, i\varepsilon\{1,2,...,N_{sta}\}$. To complete the scheduling of the STAs, the modelling of these queues is important. Let $\lambda = \{\lambda_i\}, i\varepsilon\{1,2,...,N_{sta}\}$, denote the list of arrival rates for different STAs index from 1 to N_{sta}. Figure 5.1 shows the queue pictorially. We can see that a number of queues being filled with arrival rates λ_i. The scheduler makes the decisions for the system of access points to serve one or more than one queues. Let the length of the different queues be denoted by $L = L_i$, $i\varepsilon\{1,2,...,N_{sta}\}$, where L_i is the length of queue for STA_i.

5.2 Schedule for only JT scenario

Let us first focus on a system with 2-APs and 2-STAs served using a single channel due to which they can only be served in JT mode. The 2-APs jointly serve the two STAs for some time duration in alternate manner. A **step** in the schedule is defined as serving an STA according to the the schedule. Ex - if we start by serving STA1 first, we say we took one step in the schedule when we move to serving STA2. Let us L_j^i denote the length of the queue at step i for STA_j where $j\varepsilon\{1,2\}$. We represent the initial state of the queue before service starts as L_1^0 and L_2^0.

We will assume an initial queue length $L^0 = \{L_1^0, L_2^0\}$ and serve one STA until the queue length at the start of service is served. The schedule will then move on to the other STA and serve it for a time till it's queue length at the start of service is served. The schedule is given by algorithm 1.

Remark 5.2.1 *Do not confuse that the schedule will serve an STA till the entire queue is empty. If the queue length at the start of the service is 1000 bytes, the APs will serve 1000 bytes, but meanwhile the queue length will* ***increase*** *due the new packets coming according to the arrival rate.*

Algorithm 1 Schedule for fluid model JT

Require: C^{JT}, λ
  $L_1^{curr} \leftarrow L_1^0$
  $L_2^{curr} \leftarrow L_2^0$
  $sta_to_serve \leftarrow STA_1$
  $T_{service} \leftarrow 0$
  while $True$ **do**
    if sta_to_serve is STA_1 **then**
      $T_{service} \leftarrow \frac{L_1^{curr}}{C^{JT}[1]}$ ▷ serve STA_1 till L_1^{curr} packets are served
      $L_1^{curr} \leftarrow \lambda[1] * T_{service}$
      $L_2^{curr} \leftarrow L_2^{curr} + \lambda[2] * T_{service}$
      $sta_to_serve \leftarrow STA_2$
    else if sta_to_serve is STA_2 **then**
      $T_{service} \leftarrow \frac{L_2^{curr}}{C^{JT}[2]}$ ▷ serve STA_1 till L_2^{curr} packets are served
      $L_2^{curr} \leftarrow \lambda[2] * T_{service}$
      $L_1^{curr} \leftarrow L_1^{curr} + \lambda[1] * T_{service}$
      $sta_to_serve \leftarrow STA_1$
    end if
  end while

The aim of the schedule is to make the system queue stable. The idea of a stable queue is given by definition 5.2.1.

Definition 5.2.1 ***Stable queue****: the queue is said to be stable when the queue lengths of the individual STAs do not increase indefinitely. Each of the queue lengths may approach a finite limit or could approach 0 which is the best case. Cases where the queue lengths keep repeating in a pattern without tending to infinity are also considered stable. Mathematically we can write it as equation 5.1.*

$$\lim_{i \to +\infty} L_1^i + L_2^i < \infty \tag{5.1}$$

5.3 Queue evolution modeling

Once we are introduced with the notations and the schedule used for analysing the queue, we can now try to understand the evolution of the queue lengths over time after each step in

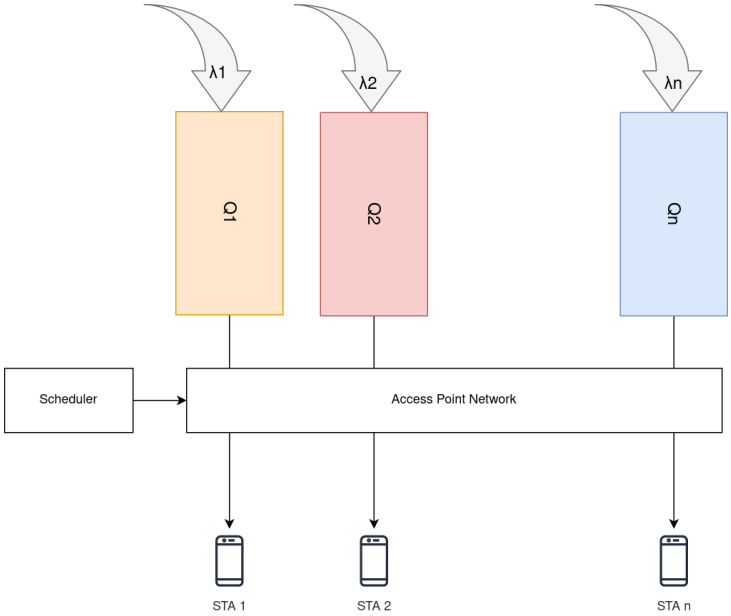


Figure 5.1: Queue Model in the system

the schedule. Taking L^1 and L_2^0 as the initial queue lengths and starting with serving STA_2, the evolution of the queue is pictorially represented in figure 5.2 and 5.3 for STA_1 and STA_2 respectively. The blue and green regions indicate STA_1 and STA_2 getting served respectively. From figure 5.2 we can observe that since the schedule serves STA_2 first, the queue length for STA_1 increases with rate λ_1 transitioning from state L_1^0 to L_1^1. During this duration, from figure 5.3 we can observe that the queue length for STA_2 decreases with rate C_2^{JT} but since there will be incoming packets while we aim to serve L_2^0 packets completely, the queue length increases with rate λ_2. The net effect on the value of queue length at the next step, L_2^1, will depend on the values of C_2^{JT} and λ_2.
In the next step, the schedule will instruct the 2-APs to serve STA_1 which will decrease the queue length with rate C_1^{JT} and at the same time increase it with rate λ_1. The net rate of decrease is represented as the slope of the line connecting L_1^1 and L_1^2 in the figure.

5.4 Modelling transitions as matrix equation

The queue lengths after one complete round of service i.e. L^2 can be written in terms of the state before 2 steps i.e. L^0 in a matrix equation. We choose to focus on the changes after one complete round of service rather than immediate changes after each service of an STA as it is easier to model the system as the equations are repetitive after both STAs are served. Assume

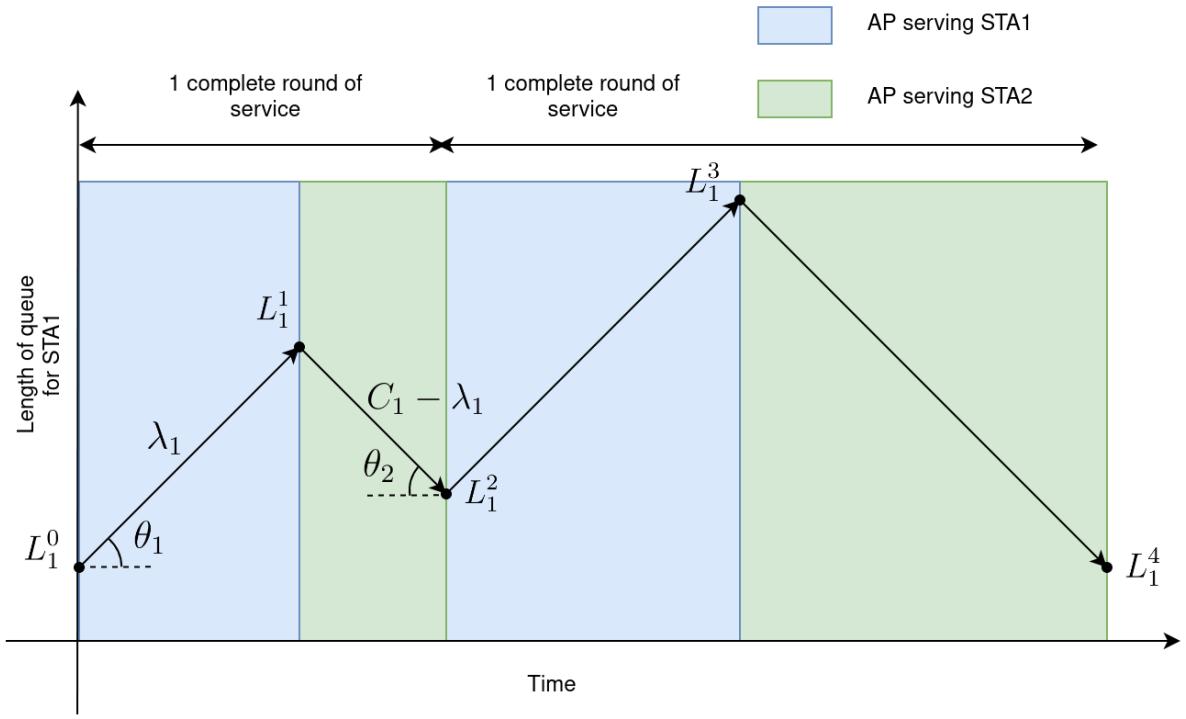


Figure 5.2: Queue evolution for STA1 assuming STA2 is served first

the following notations:

$$L^2 = \begin{pmatrix} L_1^2 \\ L_2^2 \end{pmatrix} \tag{5.2}$$

$$A = \begin{pmatrix} \frac{\lambda_1}{C_1^{JT}} + \frac{\lambda_1 \lambda_2}{C_1^{JT} C_2^{JT}} & \frac{\lambda_1}{C_2^{JT}} \\ \frac{\lambda_2^2}{C_1^{JT} C_2^{JT}} & \frac{\lambda_2}{C_2^{JT}} \end{pmatrix} \tag{5.3}$$

$$L^0 = \begin{pmatrix} L_1^0 \\ L_2^0 \end{pmatrix} \tag{5.4}$$

Then the next state after 1 complete round of service can be expressed as equation 5.5:

$$L^2 = AL^0 \tag{5.5}$$

Using the above matrix relation, we can get the values of the state of the two queues after successive complete rounds of service (i.e. at steps 2,4,6,...). The general equation representing queue lengths after i complete round of services, i is a positive integer, is given by equation 5.6:

$$L^{2i} = A^i L^0 \tag{5.6}$$

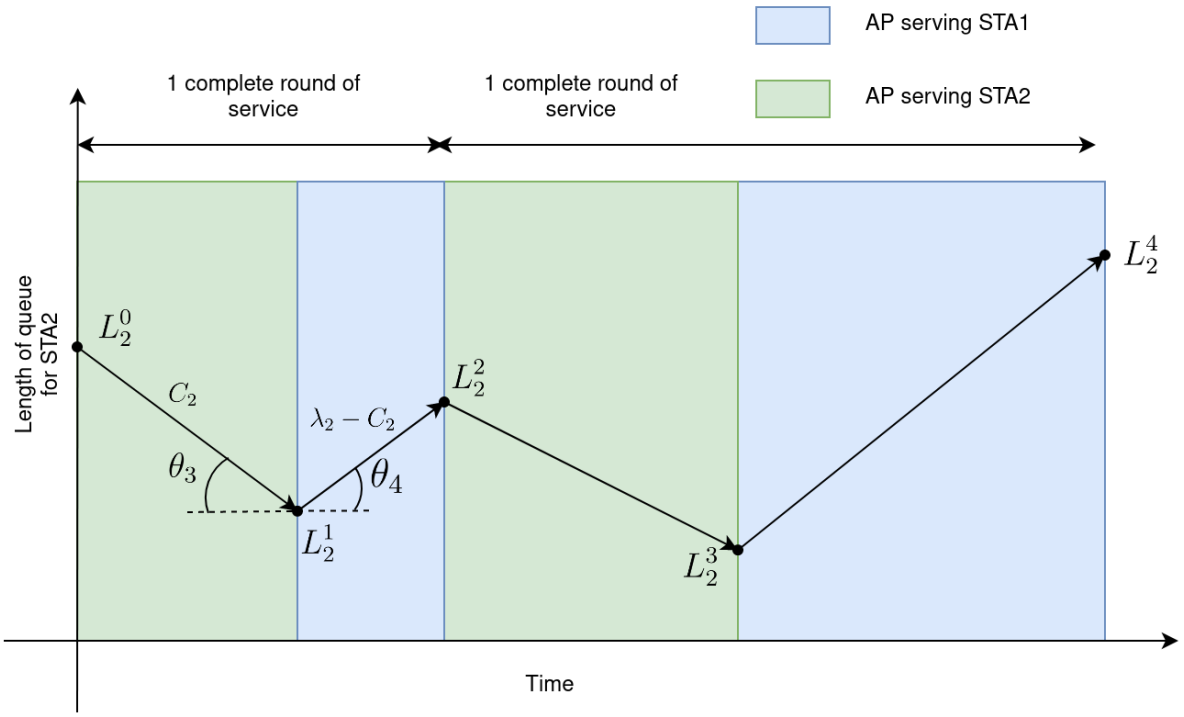


Figure 5.3: Queue evolution for STA1 assuming STA2 is served first

5.5 Stability

5.5.1 Greedy Stability

Let us assume that we want to pursue a greedy approach to serve the queue such that the total length of the queue will should decrease at each step. We need to find the relationships between the arrival rates and capacities which can satisfy the condition.

Theorem 5.5.1 *If we want the total length of the queues, in each step of serving the STA to decrease, then the necessary condition is given by equation 5.7.*

$$(\lambda_1 + \lambda_2) < \frac{C_1 + C_2}{2} \tag{5.7}$$

Proof 5.5.1 *See appendix B.*

5.6 Eigenvector analysis

5.6.1 Eigenvector equation

It can be possible that for certain conditions, the transformation brought by A results in just scaling (upscaling or downscaling) of the queue state L. Vector L for which such scaling occurs are the eigenvectors of A. We will analyse the eigenvalues and eigenvectors in this section.

It is possible that the queue lengths remain the same after a complete round of service, i.e. $L^2 = L^0$. While discussing stability of the queues, we can consider a stationary system queue as stable since the individual queue lengths for different STAs is not going to infinity as time goes by. A queue that goes to infinity is undesired and unstable as the corresponding STA will never be served properly.

The stationary property of the system queue means that the linear transformation A does not change the system queue i.e. L^0. Mathematically we can say that the eigenvalue, denoted by μ for the transformation matrix A is 1. This can be represented as equation 5.8.

$$AL^0 = \mu L^0 \tag{5.8}$$

Finding the relationships between λ and C for $\mu = 1$, we get:

$$det(A - \mu I) = 0 \tag{5.9}$$

$$\begin{pmatrix} \frac{\lambda_1}{C_1} + \frac{\lambda_1\lambda_2}{C_1C_2} - \mu & \frac{\lambda_1}{C_2} \\ \frac{\lambda_2^2}{C_1C_2} & \frac{\lambda_2}{C_2} - \mu \end{pmatrix} = 0 \tag{5.10}$$

$$C_1^2C_2^2\mu^2 - (C_1^2C_2\lambda_2 + C_1C_2^2\lambda_1 + C_1C_2\lambda_1\lambda_2)\mu + C_1C_2\lambda_1\lambda_2 = 0 \tag{5.11}$$

Assuming $C_1 > 0, C_2 > 0$, we can simplify the above equation as equation 5.12, where $\rho_i = \frac{\lambda_i}{C_i}$.

$$\mu^2 - (\rho_1 + \rho_2 + \rho_1\rho_2)\mu + \rho_1\rho_2 = 0 \tag{5.12}$$

Remark 5.6.1 *We can observe that the quadratic in μ is no longer dependent on 4 variables (arrivals and capacities), but on the ratio of the arrival to capacity of each STA. Thus reducing it to 2 variables.*

5.6.2 Eigenvalues and eigenvectors for stationary queues

Let $\mu = \{\mu_1, \mu_2\}$ denote the set of eigenvalues of the eigenvector equation 5.8. The general solution to the quadratic in μ is given by equation 5.13.

$$\mu = \frac{(\rho_1 + \rho_2 + \rho_1\rho_2) \pm \sqrt{(\rho_1 + \rho_2 + \rho_1\rho_2)^2 - 4\rho_1\rho_2}}{2} \tag{5.13}$$

The condition for one of the eigenvalues $\mu_1 = 1$ is given by the theorem 5.6.1. Using equation 5.14 and 5.13 we can say that when $\mu_1 = 1$, the other eigenvalue $\mu_2 = \rho_1\rho_2$.

Theorem 5.6.1 *If the system queue remains in the same state after one complete round of service i.e. $L^i = L^0$, where i is a positive even integer, then it must satisfy equation 5.14 assuming $C_1 > 0$ and $C_2 > 0$:*

$$\rho_1 + \rho_2 = 1 \tag{5.14}$$

Proof 5.6.1 *See appendix C.*

Theorem 5.6.2 *If one of the eigenvalues of A, μ_1, is equal to 1, then the eigenvectors corresponding to μ_1, μ_2 lie on a straight line given by the equation 5.15 and 5.16 respectively, assuming $L_1 \geq 0$ and $L_2 \geq 0$.*

$$L_{1,\mu_1} = \frac{C_1C_2 - \lambda_2C_1}{\lambda_2^2} L_{2,\mu_1} \tag{5.15}$$

$$L_{1,\mu_2} = \frac{-C_1}{C_2} L_{2,\mu_2} \tag{5.16}$$

where L_{i,μ_j} is the length of the queue of STA i for eigenvalue μ_j.

Proof 5.6.2 *See appendix D.*

Remark 5.6.2 *The eigenvectors corresponding to the eigenvalue $\mu_2 = \rho_1\rho_2$ will have a negative value of L_1, the length of queue for STA1, for positive L_2 which is not possible. Hence this eigenvector will not exist for a real queue.*

Let $D(L_1^i, L_2^i)$ be a function which gives the direction vector to the next state of queue lengths after a complete round of service if (L_1^i, L_2^i) and (L_1^{i+2}, L_2^{i+2}) are the current and the next state of the system respectively. $D(L_1^i, L_2^i)$ can be given by equation 5.18.

$$D(L_1, L_2) = x\hat{i} + y\hat{j} \tag{5.17}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} L_1^{i+2} \\ L_2^{i+2} \end{pmatrix} - \begin{pmatrix} L_1^i \\ L_2^i \end{pmatrix} \tag{5.18}$$

where x and y are the components of the direction vector in the x and y axis respectively.

Theorem 5.6.3 *If (L_1, L_2) is an eigenvector of A when $\mu_1 = 1$, then the direction at a point $(L_1 + \varepsilon, L_2 - \varepsilon)$, i.e. $D(L_1 + \varepsilon, L_2 - \varepsilon)$ given by equation 5.18, will be such that $x < 0$ and $y > 0$.*

Proof 5.6.3 *See appendix E.*

Theorem 5.6.4 *If one of the eigenvalue of A is 1, then any point (L_1, L_2) other than the eigenvectors for $\mu = 1$, will have a direction $D(L_1, L_2)$ towards the straight line representing the eigenvectors.*

Proof 5.6.4 *See appendix E.*

5.7 Jitter analysis

Jitter is a measure of the variation in the delay suffered by packets. It is defined as the expected value of the absolute difference between the delay of successive packets. We assume continuous packet transmission which means we do not assume a fixed packet size. Hence, we do not wait until sufficient data to form a packet is collected in the queue, rather we try to serve the bit as soon as it arrives. Let us denote the delay of packet arrived at time t as d_t and jitter of STA indexed i as J_i. In a stationary condition where the lengths of the queues remain constant after 1 complete round of service, the time duration of serving STA1 and STA2, denoted by $T1$ and $T2$ respectively, will remain same over time. From figure 1, assuming we start serving the

queues with serving STA2, we can see the evolution of the queue for STA1 which can be used to calcute the jitter when condition $\mu = 1$ is satisfied.

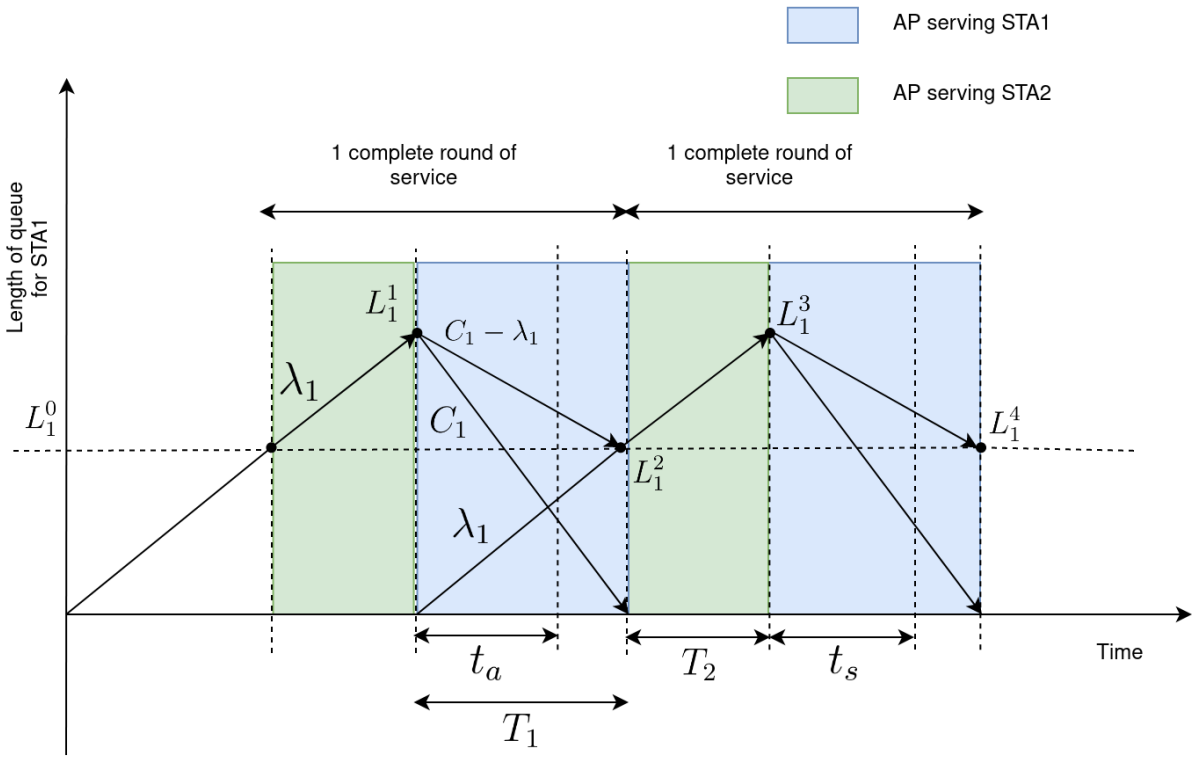


Figure 5.4: Queue evolution for STA1 if STA2 is served first for jitter analysis when $\mu = 1$

In the figure, our schedule ensures that all the packets arriving in the time duration marked as T_1 (duration when STA1 is getting served) and T_2 (duration when STA2 is getting served) will be served in the next T_1 duration. This can be mathematically written as:

$$\lambda_1(T_1 + T_2) = C_1 T_1 \tag{5.19}$$

Consider a packet that arrived at time t after we start serving STA1. This packet will be served the next time STA1 gets served. The time when this packet is served denoted by t_s in the figure, is given by using equation 5.19.

$$t_s = \frac{\lambda_1 t}{C_1} \tag{5.20}$$

We can find the delay experienced by this packet using the figure and equation 5.20. It is given as:

$$d_t = T_1 + T_2 + t_s - t \tag{5.21}$$
$$= T_1 + T_2 + (\frac{\lambda_1}{C_1} - 1)t \tag{5.22}$$

Jitter can be found of using the integration given by:

$$J_1 = \int_0^{T_1+T_2} (|\frac{d(d_t)}{dt}|)\, dt \tag{5.23}$$
$$J_1 = |(\frac{\lambda_1}{C_1} - 1)(T_1 + T_2)| \tag{5.24}$$
$$= |(\frac{T_1}{T_1 + T_2} - 1)(T_1 + T_2)| \tag{5.25}$$
$$= T_2 \tag{5.26}$$

Similarly, the jitter for STA2, denoted by J_2, is given by:

$$J_2 = T_1 \tag{5.27}$$

The jitter for different STAs for $\mu = 1$ can be found out using the equation 5.26 and 5.27. The total jitter of the system will be the average of J_1 and J_2.

$$\begin{aligned} J_{JT}|_{\mu=1} &= \frac{J_1 + J_2}{2} \\ &= \frac{T_1 + T_2}{2} \end{aligned} \tag{5.28}$$

Assuming L_1, L_2 are the stationary values of the queue, T_1 and T_2 are given by:

$$T_1 = \frac{L_1}{\lambda_1} \tag{5.29}$$
$$T_2 = \frac{L_2}{\lambda_2} \tag{5.30}$$

Using equation 5.28 and 5.30, the average jitter in JT mode using fluid model is given as:

$$J_{JT}|_{\mu=1} = \frac{1}{2}(\frac{L_1}{\lambda_1} + \frac{L_2}{\lambda_2}) \tag{5.31}$$

5.8 Results

5.8.1 Quiver plot for Fluid model

Quiver plots gives us an idea of the direction in which the system is going to move from a given state. Our 2-queue system can also be analysed by such quiver plots. Since the movements from one state to another is deterministic, such plots can be used to understand stability points in the search space. Taking $\lambda = \{10, 10\}$ KBps and $C = \{20, 20\}$ KBps such that the arrival and caQuiver plots gives us an idea of the direction in which the system is going to move from a given state. Our 2-queue system can also be analysed by such quiver plots. Since the movements from one state to another is deterministic, such plots can be used to understand stability points in the search space. Taking $\lambda = \{10KBps, 10KBps\}$ KBps and $C = \{20KBps, 20KBps\}$ KBps such that the capacities satisfy the condition in theorem 5.6.1, an example quiver plot is shown in figure 5.5. The color coding of the arrows is done in such a way that the arrows with similar directions are given similar color and the color changes gradually as the direction changes.

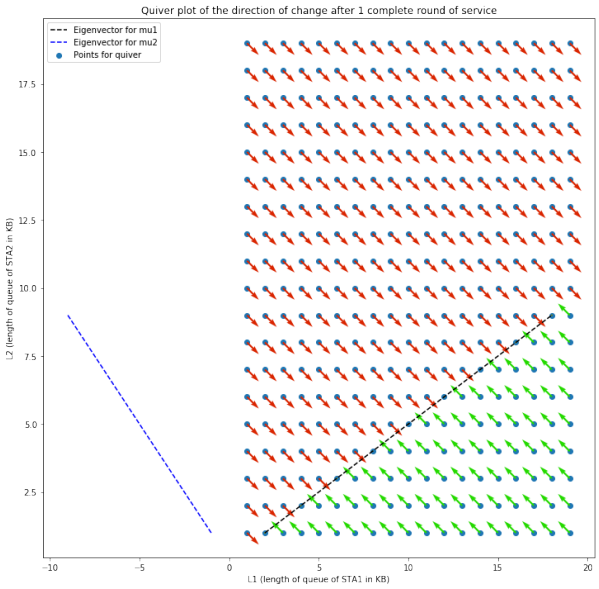


Figure 5.5: Quiver plot for condition given by Theorem 5.6.1

Observations:

1. We can see that there exist some points for which the direction is not mentioned. They can be seen as the gap between the green and red arrows. These are the points which

remain in the same state after 1 complete round of service and hence there direction vector is not-defined.

2. We can observe that these points lie on a straight line. This is in line given by Theorem 5.6.2.

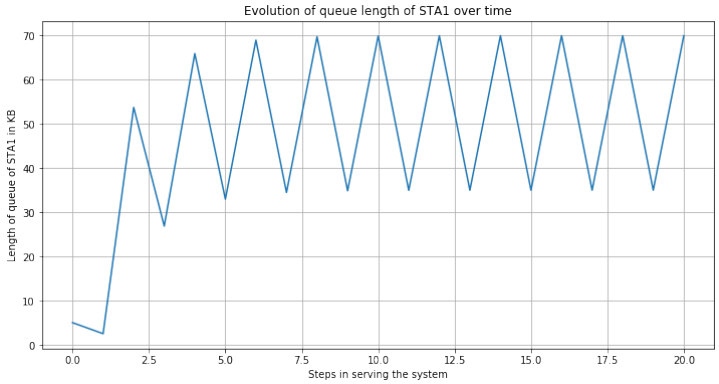
3. We can also see that the points either side of this line have a tendency to move towards this line.

4. The blue line shows the straight line corresponding to the other eigenvector for μ_2. We can see that the line does not lie in the first quadrant i.e. the points on the line do not represent real queue lengths. Hence we can discard the effect of this eigenvalue.

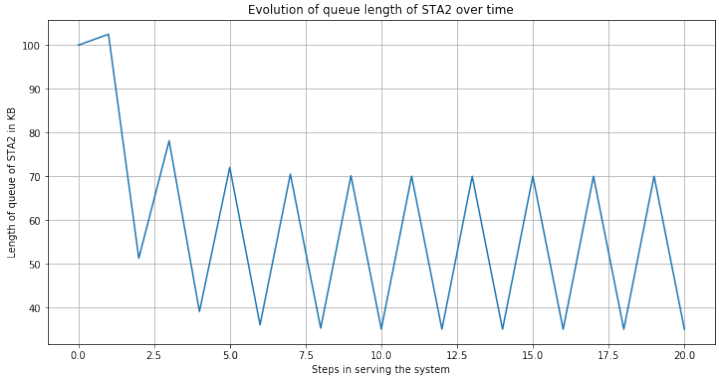
5.8.2 Simulation of queue evolution

For the same values of arrivals and capacities of the 2 STA system, and taking initial queue lengths $L = \{5, 100\}$ KB, figure 5.6 shows the simulation of evolution of queue lengths of STA1, STA2 and the total system queue for 20 steps. Each time a STA is served in the scheduling, we can refer it to as a step. We can observe that the individual queues in figure 5.6a and 5.6b tend to oscillate between just two constant states irrespective of starting with different queue lengths. Since λ and C satisfy the stationary property as in Theorem 5.6.1, we can observe that the total queue length in figure 5.6 remains constant after every 2 steps validating Theorem 5.6.1.

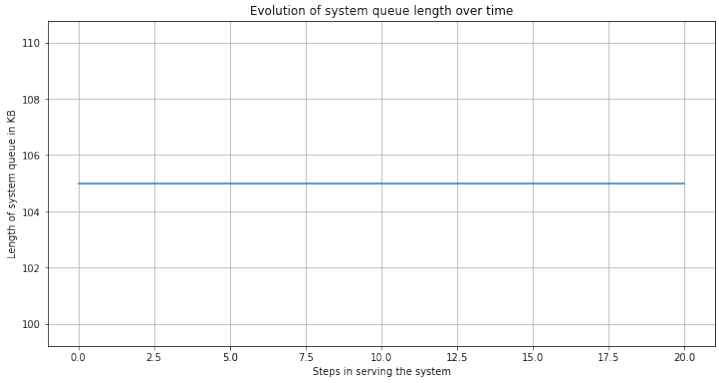
Taking arrival $\lambda = \{10, 10\}$ KBps, STA capacity $C = \{30, 30\}$ KBps and initial queue length $L = \{5, 100\}$ KB, figure 5.7 shows the simulation results representing the evolution of the individual queue lengths of the two STA and the total system queue length. From figure 5.7a and 5.7b, we can observe that the queue lengths for STA1 and STA2 goes through cycles of increase and decrease as they eventually tend to 0. As one STA1 is served, the queue length of STA2 increases and vice-versa which causes the increasing-decreasing nature of the plot. As the arrivals and STA capacity as taken according to Theorem 5.5.1, we can observe that the changes are such that the total queue length shown in figure 5.7c decreases at every step, validating Theorem 5.5.1.



(a) Queue evolution for STA1

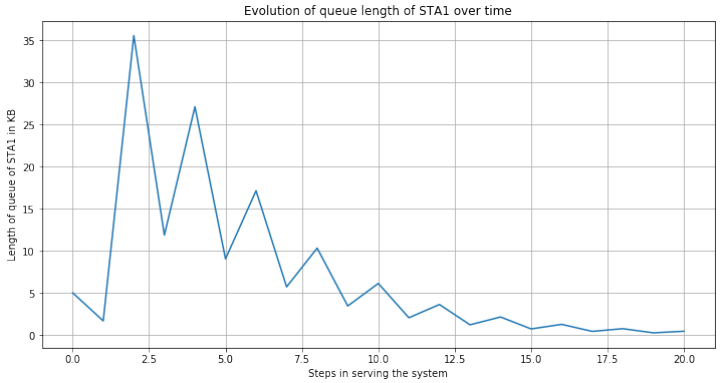


(b) Queue evolution for STA2

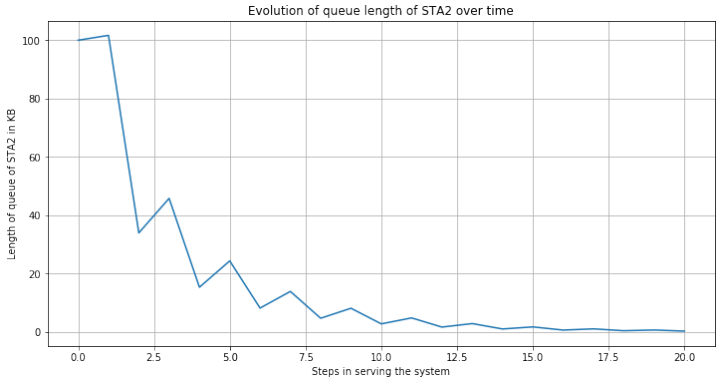


(c) Evolution of total queue length

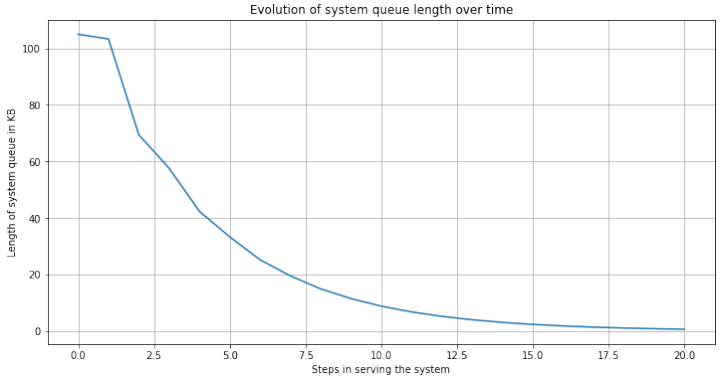
Figure 5.6: Queue evolution for STA1, STA2 and entire system over 20 steps for stationary case



(a) Queue evolution for STA1



(b) Queue evolution for STA2



(c) Evolution of total queue length

Figure 5.7: Queue evolution for STA1, STA2 and entire system over 20 steps for greedy case

CHAPTER 6

Fluid model for JT+SAPI scenario

The fluid model discussed in the previous chapter focussed on a system where the STAs were served only in JT mode. Since we need to understand a system with the options to serve both in JT and SAPI, we need to extend the fluid model to encorporate serving the STAs for some durations in JT and SAPI modes.

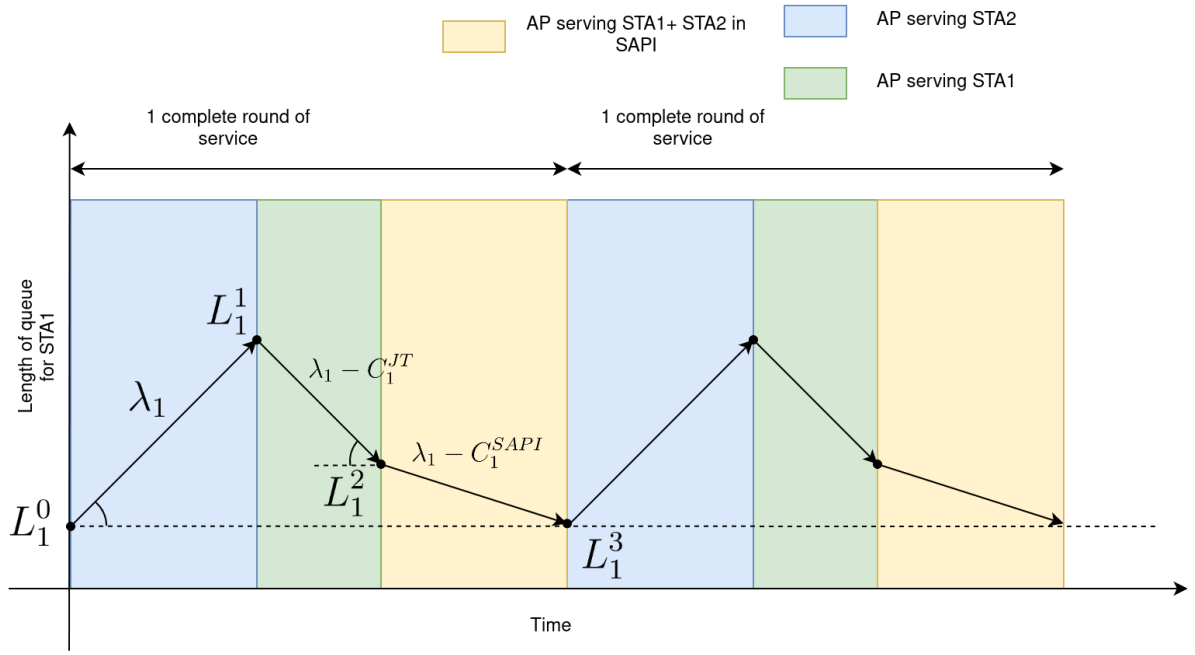


Figure 6.1: Stationary queue evolution for STA1 in Fluid model with JT+SAPI

6.1 Scheduling with JT and SAPI

Let's assume we have 2-STAs, 2-APs and 1 channel in our system. The 2-STAs will be served in such a way that STA_1 and STA_2 will be served for T_1 and T_2 in JT mode and then they will be simultaneously served in SAPI mode for a time duration of T_s.
The evolution of the queue starting from initial queue lengths, L_1^0 and L_2^0 for the two queues respectively, can be analysed for a **complete round of service**. Here a complete round of service is achieved after the STAs are served onece in JT mode and once in SAPI for some

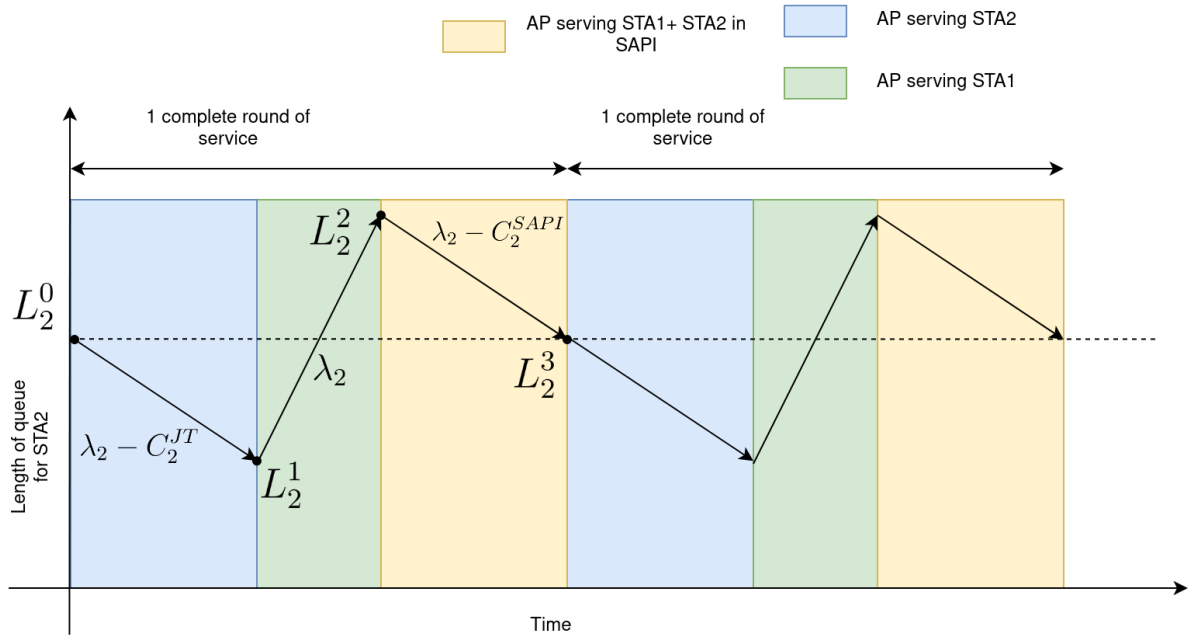


Figure 6.2: Stationary queue evolution for STA2 in Fluid model with JT+SAPI

non-zero durations. Assuming STA2 gets served first, the queue lengths after 3 steps in the schedule can be written as:

$$L_1^3 = L_1^0 + \lambda_1 T_2 + (\lambda_1 - C_1^{JT})T_1 + (\lambda_1 - C_1^{SAPI})T_s \tag{6.1}$$

$$L_2^3 = L_2^0 + (\lambda_2 - C_2^{JT})T_2 + \lambda_2 T_1 + (\lambda_2 - C_2^{SAPI})T_s \tag{6.2}$$

Remark 6.1.1 *We saw that in the case in fluid model in JT scenario, time durations T_1 and T_2 were proportional to queue lengths L_1 and L_2 respectively. This made it easy to formulate the evolution as a matrix equation. On the contrary, in the case of serving in both JT and SAPI modes, the time durations are not directly proportional to the queue lengths. Hence, we cannot write the evolution as a matrix multiplication equation.*

6.2 Stability

The queue lengths will change as the system evolves from the initial state, L^0, based on the values of T_1, T_2 and T_s. We need to choose these time durations such that the queue is stable, where we use the concept of stability as defined in equation 5.1.

One possible way of ensuring stability is to condition a situation where the queue lengths after one complete round of service remains stationary. This is represented by equation 6.3.

$$L^3 = L^0 \tag{6.3}$$

We can use this equation to find the condition for stationary property of the queue lengths.

$$(\lambda_1 - C_1^{JT})T_1 + \lambda_1 T_2 + (\lambda_1 - C_1^{SAPI})T_s = 0 \tag{6.4}$$
$$\lambda_2 T_1 + (\lambda_2 - C_2^{JT})T_2 + (\lambda_2 - C_2^{SAPI})T_s = 0 \tag{6.5}$$

We can observe that we have two equations and three variables, T_1, T_2 and T_s. Hence, there will exist an infinite number of solutions for this system of equations. Assuming some value of T_s, we can find the other variables in terms of T_s. Lets assume $\rho' = \frac{C}{\lambda}$.

Theorem 6.2.1 *For the fluid model with JT+SAPI to remain stationary after one complete round of service, the durations of service T_1 and T_2 will be given by the equations 6.6 and 6.7.*

$$T_2 = \frac{(\rho_2'^{SAPI} - \rho_1'^{SAPI}) + \rho_1'^{JT}(1 - \rho_2'^{SAPI})}{\rho'^{JT}(\rho_2'^{JT} - 1) - \rho_2'^{JT}} T_s \tag{6.6}$$
$$T_1 = (\rho_2'^{SAPI} - 1)T_s + (\rho_2'^{JT} - 1)T_2 \tag{6.7}$$

Proof 6.2.1 *We can solve equations 6.4 and 6.5 to get the equations.*

The values of T_1, T_2 obtained in equations 6.7 and 6.6 should be non-zero and positive as we want each of the STAs to be served both in JT and SAPI in a complete round of service. We can find conditions when these time durations are positive. We know that the capacity in JT will be higher than that in SAPI mode for every STA. For the system to be stable, we need a region where the net queue length is reduced. Such a condition arises while the STA is served in JT or SAPI mode. We can say that the capacity in JT mode, C^{JT}, should be greater than the arrival rate, λ.

$$C_i^{JT} > C_i^{SAPI}, i = \{1,2\} \tag{6.8}$$
$$\rho_i'^{JT} > \rho_i'^{SAPI}, i = \{1,2\} \tag{6.9}$$
$$\rho_i'^{JT} > 1, i = \{1,2\} \tag{6.10}$$

Assuming T_s is positive and focusing on conditions for T_2.

Case 1: numerator>0 and denominator>0
Simplifying the numerator, we get:

$$(\rho_2^{\prime SAPI} - \rho_1^{\prime SAPI}) + \rho_1^{\prime JT}(1 - \rho_2^{\prime SAPI}) > 0 \tag{6.11}$$

$$\rho_2^{\prime SAPI} < \frac{\rho_1^{\prime JT} - \rho_1^{\prime SAPI}}{\rho_2^{\prime JT} - 1} \tag{6.12}$$

Simplifying the denominator, we get:

$$\rho^{\prime JT}(\rho_2^{\prime JT} - 1) - \rho_2^{\prime JT} > 0 \tag{6.13}$$

$$\rho_2^{\prime JT} > \frac{\rho_1^{\prime JT}}{\rho^{\prime JT} - 1} \tag{6.14}$$

Case 2: numerator<0 and denominator<0
Simplifying the numerator, we get:

$$(\rho_2^{\prime SAPI} - \rho_1^{\prime SAPI}) + \rho_1^{\prime JT}(1 - \rho_2^{\prime SAPI}) < 0 \tag{6.15}$$

$$\rho_2^{\prime SAPI} < \frac{\rho_1^{\prime JT} - \rho_1^{\prime SAPI}}{\rho_2^{\prime JT} - 1} \tag{6.16}$$

Simplifying the denominator, we get:

$$\rho^{\prime JT}(\rho_2^{\prime JT} - 1) - \rho_2^{\prime JT} < 0 \tag{6.17}$$

$$\rho_2^{\prime JT} < \frac{\rho_1^{\prime JT}}{\rho^{\prime JT} - 1} \tag{6.18}$$

Obtaining conditions such that $T_1 > 0$ using equation 6.7, we get:

$$(\rho_2^{\prime SAPI} - 1)T_s + (\rho_2^{\prime JT} - 1)T_2 > 0 \tag{6.19}$$

We know that $\rho_2^{\prime JT} > 1$, $T_s > 0$ and $T_2 > 0$. Using these we get:

$$\rho_2'^{SAPI} > 1 \tag{6.20}$$

The queue lengths at any point in a complete round of service should not become negative as we are dealing with real length queues. The initial queue lengths with which we could start the service are conditioned ensure non-negative queue lengths. Let us assume the relation given in equation 6.21. This condition means that the queue length will be decreasing during time intervals in which the particular STA is served in JT and SAPI modes.

$$C_i^{JT} > C_i^{SAPI} > \lambda_i,\ i = \{1,2\} \tag{6.21}$$
$$\rho_i'^{JT} > \rho_i'^{SAPI} > 1,\ i = \{1,2\} \tag{6.22}$$

From the figure 6.1 we can find the condition for initial state L_1^0 for non-negative queue lengths in a complete round of service. Using equation 6.1 and condition that queue length after 1 complete round of service will be non-negative.

Theorem 6.2.2 *For a system in fluid model with JT+SAPI with time durations given by theorem 6.2.1, to have non-negative queue lengths at any point during service, the conditions on the initial queue lengths are given by equation 6.23 and 6.24.*

$$L_1^0 > (C_1^{JT} T_1 + C_1^{SAPI} T_s) - \lambda_1 (T_1 + T_2 + T_s) \tag{6.23}$$
$$L_2^0 > max((C_2^{JT} - \lambda_2) T_2, (C_2^{JT} T_2 + C_2^{SAPI} T_s) - \lambda_2 (T_1 + T_2 + T_s)) \tag{6.24}$$

Proof 6.2.2 *See appendix E.*

6.3 Delay

We assume packet based scheduling i.e. discrete packets are arrived and served unlike the previous jitter analysis where we assumed continuous packets. We can observe that the time durations, T_1 and T_2 are linear dependent on T_s, thus reducing the control parameters to just T_s. For such a simulation setup, we can analyse the delay for the different STAs for different values of T_s to understand the behaviour of the packet delay. The arrival time of the packets residing in the queue before the service begins is taken to be constant and denoted by T_{init}^{arr} which stands for arrival time of the initial packets. This parameter will influence the average

Parameter	Value
C^{JT}	$\{20, 20\}$ KBps
C^{SAPI}	$\{18, 18\}$ KBps
λ	$\{15, 15\}$ KBps
L^0	$\{50, 20\}$ KB

Table 6.1: Parameter Values in simulation for Delay analysis in JT+SAPI

delay and corresponding jitter but if averaged over long time, the effect will be negligible. For the simulation, we choose $T_{init}^{arr} = -5$s, which means the packets have arrived 5 seconds prior to the start of the service (origin is considered at the start of the service).

Using the values of the arrival, capacities and initial values given in Table 6.1, we obtain plots for delay of STA1 and STA2 as shown in figure 6.3a and 6.3b for $T_s = 5$s and $T_s = 10$s respectively.

Observations and explanations:

1. We can observe that the delay plot shows a periodic pattern as the packets are served. This nature is common for both $T_s = 5$s and $T_s = 10$s.

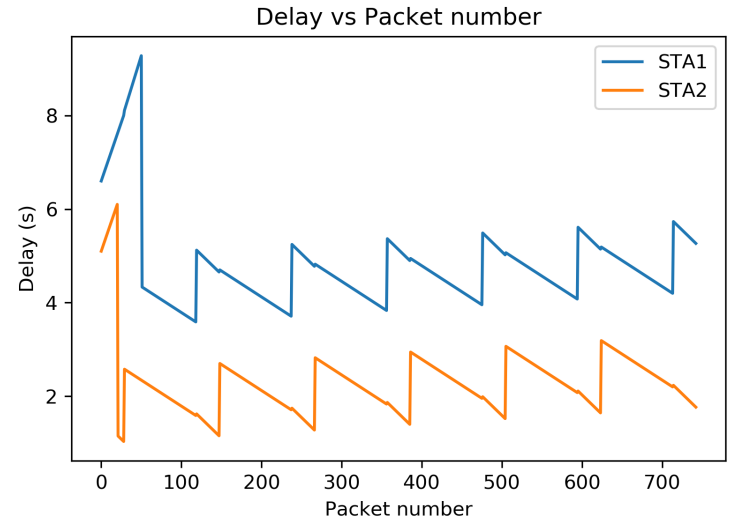
2. The periodic pattern can be attributed to the fact that the service is under the condition that the queue length is stationary after each complete round of service.

3. We can observe that the delay is relatively higher for the starting packets as compared to the successive packets. These are the initial packets in the queue and the value of T_{init}^{arr} causes the different behaviour that the rest of the delay plot.

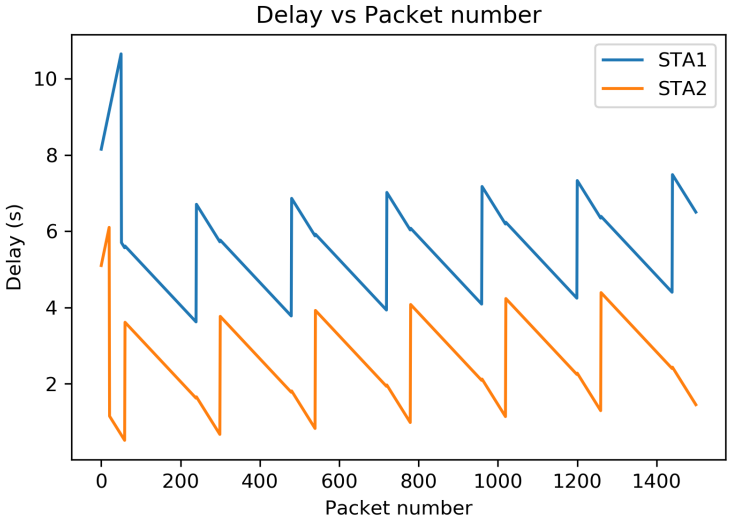
4. In the periodic nature of the delay, we can see sudden jumps in an otherwise decreasing delay. The packets that come in the duration when the corresponding STA is being served, not all of them are served in this duration. The packets which are left to be served later have significantly more delay causing the sudden jumps.

6.4 Jitter

The jitter analysis for such a system where STAs are served both in JT and SAPI mode can be done to understand the variation of jitter over system parameters. Average Jitter plot as T_s increases is shown in figure 6.4 using the values of arrivals, capacities and other parameters as given in table 6.2. The values of initial queue lengths are taken such that they lie in the permissible value range given by conditions in equation E.11 and E.17. The system was simulated for 10 complete rounds of service.



(a) Delay plot for simulation with $T_s = 5$ s



(b) Delay plot for simulation with $T_s = 10$ s

Figure 6.3: Delay Vs Packet Number for simulation of Fluid model with JT+SAPI

Parameter	**Value**
C^{JT}	$\{20, 20\}$ KBps
C^{SAPI}	$\{18, 18\}$ KBps
λ	$\{15, 15\}$ KBps
L^0	$\{51, 200\}$ KB
T_s	$(1, 100)$ s

Table 6.2: Parameter Values in simulation for Jitter analysis in JT+SAPI

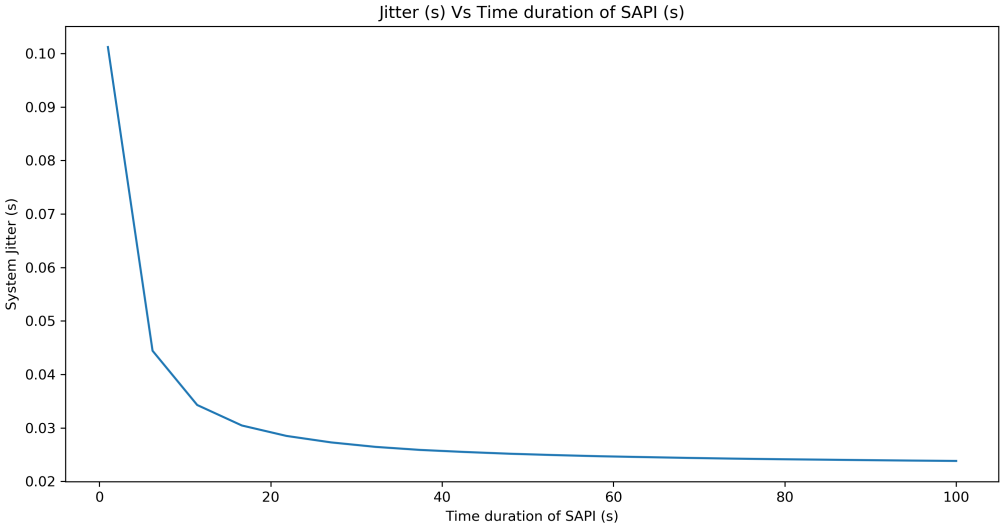


Figure 6.4: Average Jitter Vs T_s

Observations and explanations:

1. The average jitter faced by the served packets decreases as we increase T_s (time duration of serving in SAPI).

2. The frequency of the sudden jumps in the delay for packets (as discussed in the delay plot observations) influences the jitter. More frequent jumps will increase the jitter as the variation of delay will be higher. As T_s is increased, the frequency of such jumps reduce as more packets can be served since the duration of service is larger. This is the reason behind the decrease in average jitter of the packets.

CHAPTER 7

Conclusion and Future Scope

RIS-Assisted Wireless Networks The analysis presented in the thesis on RIS-assisted wireless networks in noise limited scenario consisted of received SNR variations with system parameters. It was found that the improvement over no-RIS case, for feasible RIS panel deployment, was negligible when the LoS path signal propagation followed free space path loss model $\alpha = 2$. On the other hand, the improvement increased tremendously if the LoS path contained some scatterers. The specific case considered in the analysis was $\alpha = 4$. We observed that the SNR was near linear in capture angle ($\bar{\theta}$) and showed a knee point characteristic behaviour as capture radius (r) increases. The observed increase in SNR in mobile scenario strengthened the claim that using RIS will be useful.

The knee point analysis gave us some insights that there exist certain regions around which maximum improvement can be achieved by deploying less number of RIS panels. Such regions (knee points) can be beneficial in allocation of RIS elements which is the next challenge to be solved if we want to design and analyse the network at a system level. This can be taken up as future work on the topic. We can also include more complicated indoor and outdoor path loss models to get near to the practical situations where the RIS will be deployed.

Multi-AP Coordination The work on Multi-AP Coordination presented in the thesis highlights the usefulness of the concept in terms of showing the fraction of the total STAs which can be benefitted if served in JT. The concept of choosing between JT and SAPI modes was introduced along with the idea of using modulation coding schemes. An ellaborate discussion on the queue evolution in the fluid model was presented with proposed conditions for stability and stationary queue lengths. Mathematical modelling using eigenvector equations was accomplished which helped ease the analyse of the various conditions presented in the form of theorems. The fluid model was analysed for only JT mode and was extended to JT+SAPI mode services. The stability and stationary queue conditions for the later was also presented. Mathematical simulations supported the proposed conditions.

The system was analysed on a small scale consisting of only 2 STAs. As part of future work, this constraint could be relaxed and a general mathematical model for serving STAs in JT or JT+SAPI mode could be worked upon. The capacity calculations assumed free space path loss

model which can be extended to include more complex channel fading. We can also work on algorithms that gives us a schedule which could incorporate the fact that capacities of the STAs change over time.

CHAPTER A

Proof of Theorem 2.5.1

Using the linearity properties, we have,

$$S(\bar{\theta}, \rho) = k\bar{\theta}\rho + S_0, \tag{A.1}$$

where, S_0 is SNR of network without RIS aid, i.e. $\rho = 0$. Substituting the parameters of A and B in (A.1), letting $S(\bar{\theta}_A, \rho_A) = S(\bar{\theta}_B, \rho_B)$ and by algebraic manipulations, we get,

$$\rho_A = \rho_B \bar{\theta}_B / \bar{\theta}_A. \tag{A.2}$$

Since all the parameters are non zero values, a finite ρ_A exists.

CHAPTER B

Proof of Theorem 5.5.1

Let us assume that the STA1 is getting served in step 1. The length of the 2 queues after 1 step will look like:

$$\begin{pmatrix} L_1^1 \\ L_2^1 \end{pmatrix} = \begin{pmatrix} \frac{\lambda_1 L_1^0}{C_1} \\ L_2^0 + \frac{\lambda_2 L_1^0}{C_1} \end{pmatrix} \tag{B.1}$$

The queue lengths after another step, when STA2 is getting served, will look like:

$$\begin{pmatrix} L_1^2 \\ L_2^2 \end{pmatrix} = \begin{pmatrix} L_1^1 + \frac{\lambda_1 L_2^1}{C_2} \\ \frac{\lambda_2 L_2^1}{C_2} \end{pmatrix} \tag{B.2}$$

The condition that the total length of queue should decrease from step 0 to step 1 gives us the relation:

$$L_1^0 + L_2^0 > L_1^1 + L_2^1 \tag{B.3}$$

$$L_1^0 + L_2^0 > \frac{\lambda_1 L_1^0}{C_1} + (L_2^0 + \frac{\lambda_2 L_1^0}{C_1})$$

$$L_1^0 > (\frac{\lambda_1}{C_1} + \frac{\lambda_2}{C_1}) L_1^0$$

$$(\frac{\lambda_1 + \lambda_2}{C_1}) < 1 \tag{B.4}$$

Similarly, the condition that the total queue length should decrease from step 1 to step 2 can give us:

$$L_1^1 + L_2^1 > L_1^2 + L_2^2 \tag{B.5}$$

$$L_1^1 + L_2^1 > (L_1^1 + \frac{\lambda_1 L_2^1}{C_2}) + \frac{\lambda_2 L_2^1}{C_2}$$

$$L_2^1 > (\frac{\lambda_1}{C_2} + \frac{\lambda_2}{C_2}) L_2^1$$

$$(\frac{\lambda_1 + \lambda_2}{C_2}) < 1 \tag{B.6}$$

Combining equations B.6 and B.4, we get 5.7.

CHAPTER C

Proof of Theorem 5.6.1

Using $\mu = 1$ in equation 5.11, we get:

$$C_1^2 + C_2^2 = C_1^2 C_2 \lambda_2 + C_1 C_2^2 \lambda_2 \tag{C.1}$$

$$C_1 C_2 = C_1 \lambda_2 + C_2 \lambda_1 \tag{C.2}$$

$$\frac{\lambda_1}{C_1} + \frac{\lambda_2}{C_2} = 1 \tag{C.3}$$

$$\rho_1 + \rho_2 = 1 \tag{C.4}$$

CHAPTER D

Proof of Theorem 5.6.2

The proof can be done by solving the matrix equation $AL = L$ and showing that the two equations obtained in L_1, L_2 are not independent using the theorem 5.6.1. For $\mu_1 = 1$, we get:

$$AL = 1L$$

$$(\frac{\lambda_1}{C_1} + \frac{\lambda_1\lambda_2}{C_1C_2} - 1)L_1 + (\frac{\lambda_1}{C_2})L_2 = 0 \tag{D.1}$$

$$(\frac{\lambda_2^2}{C_1C_2})L_1 + (\frac{\lambda_2}{C_2} - 1)L_2 = 0 \tag{D.2}$$

We can use theorem 5.6.1 to prove that equation D.1 and D.2 represent the same equation and hence have infinite solutions. Simplifying equation D.1:

$$\begin{aligned}
((\frac{\lambda_1}{C_1} - 1) + \frac{\lambda_1\lambda_2}{C_1C_2})L_1 + (\frac{\lambda_1}{C_2})L_2 &= 0 \\
((\frac{\lambda_2}{C_2})(\frac{\lambda_1}{C_1} - 1)L_1) + (\frac{\lambda_1}{C_2})L_2 &= 0 \\
(\frac{-C_2}{C_1})((\frac{-\lambda_2^2}{C_2^2})L_1 + (\frac{\lambda_1}{C_2})L_2) &= 0 \\
(\frac{\lambda_2^2}{C_1C_2})L_1 + (\frac{\lambda_2}{C_2} - 1)L_2 &= 0
\end{aligned}$$

which is equation D.1. The straight line equation given by equation D.2 represents the eigenvectors for $\mu_1 = 1$.

We can similarly prove that the set of equations from $AL = (\frac{\lambda_1\lambda_2}{C_1C_2})L$ also represent the same equation and get the straight line representing eigenvectors for $\mu_2 = \frac{\lambda_1\lambda_2}{C_1C_2}$.

CHAPTER E

Proof of Theorem 5.6.3

Let us assume that $(L_1^0, L_2^0) = (L_1 + \varepsilon, L_2 - \varepsilon)$ and the next point after 1 complete round of service be denoted by (L_1^2, L_2^2). This can be represented in matrix form as:

$$\begin{pmatrix} L_1^0 \\ L_2^0 \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} + \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} \tag{E.1}$$

Using equation 5.5 and E.1 we can write that:

$$\begin{pmatrix} L_1^2 \\ L_2^2 \end{pmatrix} = A \begin{pmatrix} L_1^0 \\ L_2^0 \end{pmatrix} \tag{E.2}$$

$$= A \begin{pmatrix} L_1 + \varepsilon \\ L_2 - \varepsilon \end{pmatrix} \tag{E.3}$$

$$= A \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} + A \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} \tag{E.4}$$

$$= 1 \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} + A \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} \tag{E.5}$$

Using equation E.5, we can write the value of $D(L_1^0, L_2^0)$ as:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} L_1^2 \\ L_2^2 \end{pmatrix} - \begin{pmatrix} L_1^0 \\ L_2^0 \end{pmatrix} \tag{E.6}$$

$$= \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} + A \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} - \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} - \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} \tag{E.7}$$

$$= (A - I) \begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix} \tag{E.8}$$

We know that the condition to get eigenvalue $\mu_1 = 1$ is given by equation 5.14. Using equation

E.8 and 5.14, we can calculate x and y as :

$$x = (\frac{-\lambda_2^2}{C_2^2} - \frac{\lambda_1}{C_1})\varepsilon$$
$$y = (\frac{\lambda_2^2}{C_1 C_2} + \frac{\lambda_1}{C_1})\varepsilon \tag{E.9}$$

We can see that for a $\varepsilon > 0$, x<0 and y>0 assuming non zero (C_1, C_2).

Proof of Theorem 5.6.4 Using theorem 5.6.3, we know that for a eigenvector (L_1, L_2) using $\varepsilon > 0$, we have proved that the point $(L_1^0, L_2^0) = (L_1 + \varepsilon, L_2 - \varepsilon)$ will have a direction vector, $D(L_1^0, L_2^0)$, pointing towards the line. Without loss of generality, as we vary ε, we have proved that for all points on the line passing through (L_1, L_2), the points will have a direction towards the straight line. This means the direction vector will point towards the second quadrant if we assume origin at the current point itself. As there are infinite points on the eigenvector line and we can say the same for any of these points, we deduce that any point below the line will be attracted to the line.
Similarly, if we take $\varepsilon < 0$ we get that the points above the straight line have a direction pointing towards the fourth quadrant, if the current state if taken origin. As there are infinite points on the eigenvector line and we can say the same for any of these points, we deduce that any point above the line will be attracted to the line.

Using the above arguments, we have proved that all the points that are not an eigenvector of A will have a direction vector pointing towards the straight line for eigenvectors of $\mu = 1$.

Proof of Theorem 6.2.1 Using figure 6.1, the queue length for STA_1 after one complete round of service should be positive. This is written as:

$$L_1^3 > 0 \tag{E.10}$$
$$L_1^0 > (C_1^{JT} T_1 + C_1^{SAPI} T_s) - \lambda_1 (T_1 + T_2 + T_s) \tag{E.11}$$

Similarly, using figure 6.2 we can find the conditions for L_2^0.

$$L_2^1 = L_2^0 + (\lambda_2 - C_2^{JT}) T_2 \tag{E.12}$$
$$L_2^1 > 0 \tag{E.13}$$
$$L_2^0 > (C_2^{JT} - \lambda_2) T_2 \tag{E.14}$$

also, the queue length after 3 steps, L_2^3, should also be positive. Using equation 6.2 we can say:

$$L_2^3 > 0 \tag{E.15}$$

$$L_2^0 > (C_2^{JT} T_2 + C_2^{SAPI} T_s) - \lambda_2 (T_1 + T_2 + T_s) \tag{E.16}$$

Using equations E.14 and E.16, we can say:

$$L_2^0 > max((C_2^{JT} - \lambda_2) T_2, (C_2^{JT} T_2 + C_2^{SAPI} T_s) - \lambda_2 (T_1 + T_2 + T_s)) \tag{E.17}$$

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